

**A
PROJECT REPORT
ON
A STUDY ON RISK AND RETURN ANALYSIS IN EQUITIES
AT
VENTURA SECURITIES LTD**

SUBMITTED

To

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY-HYDERABAD

IN PARTIAL FULFILLMENT FOR THE AWARD OF THE DEGREE OF

MASTER OF BUSINESS ADMINISTRATION

By

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UNDER THE GUIDANCE OF

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(AUTONOMOUS)

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CERTIFICATE

This is to certify that **Ms. HOSUR BINDU MANASA** bearing the roll No.: **24R21E0053** has successfully completed the project on “**A Study on Risk and Return Analysis in Equities**” towards the partial fulfilment for the award of the degree of Master of Business Administration with the specialization in FINANCE for the academic year 2025-26 and submitted to MLR Institute of Technology, Hyderabad.

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Kyon ki bhaiya, sabse bada rupaiya.

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Certificate

This is to certify that **Ms. Hosur Bindu Manasa**, a student of Master of Business Administration (MBA), has successfully completed her project with **Ventura Securities Ltd.** on the project titled "**Risk and Return Analysis in Equities**" under our guidance and supervision as part of the MBA curriculum.

During the project period, she worked sincerely and diligently. We found her performance satisfactory.

We wish her success in all her future endeavours.

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DECLARATION

I hereby declare that project report titled “**A Study on Risk and Return Analysis in Equities**” submitted by me to the department of Master of Business Administration, **MLR Institute of Technology, Hyderabad** is a Bonafide work undertaken by me and it is not submitted either in full or in part to any other University or Institution for the award of any degree diploma / certificate or published any time before.

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ABSTRACT

The present study titled "Risk and Return Analysis in Equities – A Study with Reference to Ventura Securities Ltd." aims to examine the relationship between risk and return in equity investments by analysing the performance of selected companies, namely Tata Consultancy Services (TCS), ICICI Bank Ltd., and Reliance Industries Ltd. The study is based on secondary data collected from sources such as Yahoo Finance, annual reports, and financial websites. Various financial and statistical tools, including average return, standard deviation, covariance, correlation, beta, Sharpe ratio, and financial ratio analysis, were used to evaluate the return potential, volatility, market sensitivity, and financial performance of the selected equities. The findings reveal that ICICI Bank generated comparatively higher returns with moderate risk, whereas TCS and Reliance Industries exhibited greater fluctuations during the study period. The study highlights that investment decisions should not be based solely on return expectations, as risk plays an equally important role in portfolio performance. It further emphasizes the significance of diversification, market analysis, and informed decision-making in achieving long-term financial growth and stability. The study also demonstrates the role of Ventura Securities Ltd. in facilitating equity investments by providing investors with trading services, market insights, and investment support.

CHAPTER - I

INTRODUCTION

1. INTRODUCTION AND OVERVIEW

1.1 INTRODUCTION

Equity investment is one of the most dynamic avenues in modern finance, offering investors the opportunity to participate directly in corporate growth. By holding shares, individuals and institutions gain ownership rights and the potential to earn through dividends and capital appreciation. Unlike fixed-income securities, however, equity returns are not guaranteed. They are shaped by market conditions, industry performance, and company-specific developments, making equities both attractive and risky.

Risk in equities refers to the unpredictability of returns. It can be divided into **systematic risk**, which arises from market-wide factors such as inflation, interest rates, or global crises and cannot be eliminated through diversification, and **unsystematic risk**, which is specific to individual companies and can be reduced by holding a diversified portfolio. Return, on the other hand, represents the financial gain or loss from an investment, derived from both price appreciation and dividend income. Evaluating risk alongside return allows investors to judge whether the rewards justify the uncertainty.

Risk and return are inseparable elements of investment decisions. The general principle is that when investors take on higher levels of risk, they expect the possibility of higher rewards. Conversely, investments with lower risk usually provide steadier but smaller returns. This trade-off forms the foundation of portfolio management, as every investor must decide how much uncertainty they are willing to accept in pursuit of gains.

Ultimately, risk-return analysis is not just a theoretical exercise; it is a practical framework that connects financial concepts with real market behavior. By applying this approach, investors can make more structured, evidence-based decisions that align with their financial goals and tolerance for uncertainty.

Meaning of Risk and Return Analysis in Equities

Risk and return analysis in equities refers to the study of how much investors can potentially earn from stock investments compared to the uncertainty involved in achieving those

earnings. It focuses on the variability of returns over time and the extent of price fluctuations in the market.

This analysis helps in understanding how securities behave under different economic and market conditions, and how sensitive they are to changes in the broader environment. By applying statistical methods, investors can measure performance, compare different stocks, and evaluate whether the expected return justifies the level of risk taken.

In essence, risk and return analysis provides a structured framework for making rational investment decisions in equity markets. It enables investors to balance profitability with stability and serves as a foundation for building efficient portfolios.

Objectives of Risk and Return Analysis in Equities

The objective of risk and return analysis in equities is to study the performance of selected stocks by examining both the returns generated and the level of uncertainty associated with them. It aims to analyze how variations in risk influence the returns earned from different securities. The study also focuses on comparing multiple stocks to identify those that offer relatively better returns for the degree of risk involved. Another objective is to observe the extent of price fluctuations and to understand how individual stocks respond to overall market changes. In addition, the analysis supports the development of a diversified portfolio by identifying suitable combinations of securities. Ultimately, it provides a structured approach for making sound investment decisions in the equity market, particularly with reference to Ventura Securities Ltd.

Significance of the Study

1. **Investor Perspective:** The study equips investors with practical knowledge of how risk and return interact in equity markets. It helps them identify stable stocks, recognize volatile ones, and make informed decisions that align with their financial goals and risk tolerance.
2. **Corporate Perspective (Ventura Securities Ltd.):** The findings strengthen advisory and portfolio management services by providing evidence-based insights. Ventura can use the results to design diversified portfolios and guide clients toward strategies that balance safety with growth.

3. **Academic Perspective:** The research bridges theoretical finance concepts with real market data. It demonstrates the application of statistical tools such as standard deviation, beta, correlation, and covariance, making the study valuable for academic learning and future research.
4. **Market Understanding:** The analysis highlights how selected companies respond to broader market movements, offering a clearer view of systematic and unsystematic risks. This contributes to a deeper understanding of equity market dynamics in the Indian context.

Methods of Preparing Risk and Return Analysis in Equities

- Risk and return analysis in equities is conducted through the application of quantitative methods to evaluate the performance of securities.
- The rate of return is calculated to measure how stock prices change over a specified period.
- Average return is used to understand the overall earnings trend of a particular stock.
- Standard deviation is applied to assess the degree of variability in returns, which reflects the level of risk.
- Beta is used to examine the extent to which a stock responds to fluctuations in the broader market.
- Correlation analysis helps in determining the strength and direction of the relationship between different securities or between a stock and the market index.
- These techniques together provide a comprehensive view of the risk and return characteristics of stocks.
- They also assist investors in making rational investment decisions and constructing efficient portfolios, especially with reference to Ventura Securities Ltd.

Advantages of Preparing Risk and Return Analysis in Equities

- It offers a systematic way to examine stock performance by considering both expected returns and the degree of uncertainty involved, leading to more reliable investment decisions.
- It helps investors distinguish between different securities by analyzing their risk-return characteristics, enabling better selection of investment opportunities.

- It enhances understanding of market movements and price variations, allowing investors to respond more effectively to changing market conditions.
- It aids in constructing a balanced and diversified portfolio, which can reduce overall investment risk while improving the potential for consistent returns.

1.2 NEED OF THE STUDY

Equity markets are among the most volatile segments of the financial system, constantly influenced by macroeconomic changes, government policies, technological shifts, and global events. This volatility creates uncertainty in returns, making it difficult for investors to predict outcomes with confidence. For those seeking to maximize wealth while controlling risk, a structured risk–return analysis becomes indispensable.

The study is particularly important because it evaluates how selected companies **Tata Consultancy Services, ICICI Bank, and Reliance Industries** performed over a five-year period. By applying statistical tools such as average return, standard deviation, beta, covariance, and correlation, the research highlights both profitability and variability in returns. This dual perspective allows investors to identify stocks that provide consistent performance, those that carry higher volatility, and the extent to which they move in relation to overall market trends.

For Ventura Securities Ltd., the analysis is directly relevant to its advisory and portfolio management functions. Investors want to know which stocks are safe, which are risky, and how they behave compared to the overall market. This study helps by:

- ✓ **Finding stable stocks:** showing which companies give steady returns without too much fluctuation.
- ✓ **Measuring risk:** explaining how much uncertainty or ups and downs are linked to each investment.
- ✓ **Helping diversification:** showing how combining different stocks can reduce overall risk.
- ✓ **Comparing with the market:** checking how these companies perform against the Nifty 50 index, which represents the broader market.

By doing this, the study makes it easier for Ventura Securities Ltd. to guide investors in choosing the right mix of stocks that balance safety and growth.

1.3 SCOPE OF THE STUDY

The present study focuses on examining the risk and return characteristics of selected equity shares. The scope of the study is restricted to three companies, namely, Tata Consultancy Services, ICICI Bank Limited, and Reliance Industries Limited. The analysis is carried out using historical data covering a period of five years, from April 2021 to March 2026. The stock price data required for the study has been collected from Yahoo Finance. Returns are computed based on the yearly opening and closing prices of the selected stocks. The level of risk associated with each stock is measured using statistical tools such as standard deviation. In addition, the relationship among the selected stocks is analysed with the help of covariance and correlation techniques. The study also incorporates the calculation of beta to assess the sensitivity of individual stock returns in relation to the market index, namely the Nifty 50. Furthermore, apart from analysing risk and return, the financial position of the selected companies is evaluated using important financial ratios such as Earnings Per Share (EPS), Return on Equity (ROE), Price-to-Earnings (P/E) Ratio, and Debt-Equity Ratio. Overall, the study provides a comprehensive understanding of the market performance of the selected stocks and assists investors in making informed investment decisions based on risk and return considerations.

1.4 OBJECTIVES OF THE STUDY

1. To measure the annual returns of the selected companies (TCS, ICICI Bank, Reliance) over five years.
2. To calculate the level of risk for each stock using statistical tools like variance and standard deviation.
3. To compare the performance of the three companies in terms of both profitability and volatility.
4. To examine how sensitive each stock is to overall market changes through beta analysis.
5. To evaluate the financial performance of selected companies using financial ratios such as EPS, ROE, P/E Ratio, Debt-Equity Ratio, and Sharpe Ratio.

1.5 RESEARCH METHODOLOGY

1. The study follows an analytical approach to examine the risk and return of selected equity shares.
2. It is based on secondary data obtained from reliable sources such as Yahoo Finance.
3. The period of analysis covers five years, from April 2021 to March 2026.
4. Three companies are selected for the study: Tata Consultancy Services, ICICI Bank Limited, and Reliance Industries Ltd.

Formulae Used:

- ✓ **Return Calculation:** Annual returns are computed using opening and closing stock prices:

$$\text{Return} = \frac{(P_1 - P_0)}{P_0} \times 100$$

Where P_0 is the opening price and P_1 is the closing price.

- ✓ **Risk Measurement:** The degree of risk is evaluated using variance and standard deviation:

$$\text{Variance} = \frac{\sum (R - \bar{R})^2}{n}$$

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

- ✓ **Inter-relationship Analysis:** The association between stock returns is examined using covariance and correlation:

$$\text{Covariance} = \frac{\sum [(R_x - \bar{R}_x)(R_y - \bar{R}_y)]}{n}$$

$$\text{Correlation (r)} = \frac{\text{Covariance}}{(\sigma_x \times \sigma_y)}$$

- ✓ **Market Sensitivity:** Beta is calculated to determine the sensitivity of each stock in relation to the market index (Nifty 50):

$$\text{Beta } (\beta) = \frac{\text{Covariance (Stock, Market)}}{\text{Variance (Market)}}$$

- ✓ **Sharpe Ratio:**

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

✓ **Earnings Per Share (EPS):**

$$\text{EPS} = \frac{\text{Net profit} - \text{Preference Dividend}}{\text{Number of Equity Shares}}$$

✓ **Return on Equity (ROE):**

$$\text{ROE} = \frac{\text{Net Income}}{\text{Shareholder's Equity}} \times 100$$

✓ **Price Earnings Ratio (P/E Ratio):**

$$\text{P/E Ratio} = \frac{\text{Market Price per Share}}{\text{EPS}}$$

✓ **Debt-Equity Ratio:**

$$\text{Debt-Equity Ratio} = \frac{\text{Total Debt}}{\text{Shareholder's Equity}}$$

✓ **Current Ratio:**

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

✓ **Net Profit Margin:**

$$\text{Net Profit Margin} = \frac{\text{Net Profit}}{\text{Revenue}}$$

The collected data was systematically arranged, analyzed, and interpreted to draw meaningful conclusions about the risk–return relationship. This methodology ensures that the findings are objective, quantifiable, and relevant for portfolio decision-making at Ventura Securities Ltd.

1.6 LIMITATIONS OF THE STUDY

1. The study is limited only to the selected companies considered for analysis.
2. The study is based entirely on secondary data collected from financial websites and company annual reports.
3. The analysis is restricted to the selected study period from April 2021 to March 2026. Stock market performance may change due to fluctuations in economic and market conditions.
4. External factors such as political, economic, and global market conditions are not fully considered in the study.

CHAPTER – II

REVIEW

OF

LITERATURE

2. REVIEW OF LITERATURE

2.1 Theoretical Framework

RISK ANALYSIS IN EQUITY MARKETS

Risk analysis in equity markets refers to the structured process of identifying, evaluating, and measuring the uncertainty associated with returns from stock investments. It is a critical component of investment decision-making because equity markets are highly volatile and influenced by multiple internal and external factors. The primary objective of risk analysis is to understand the degree of variability in stock returns and to determine whether the expected return adequately compensates for the level of risk undertaken.

In equity investments, risk arises because future stock prices and returns cannot be predicted with certainty. Therefore, investors rely on historical data, statistical tools, and financial models to assess risk and make informed investment decisions.

Risk analysis is not only focused on minimizing losses but also on optimizing the risk-return trade-off. A well-conducted risk analysis helps investors construct efficient portfolios, reduce uncertainty, and improve investment performance.

RISK IN EQUITY MARKETS

In finance, risk refers to the uncertainty or variability of returns associated with an investment. Markowitz (1952) introduced Modern Portfolio Theory, in which risk is defined as the variance or standard deviation of portfolio returns, measuring the dispersion of actual returns from expected returns. Equity markets are particularly exposed to risk because future stock prices are influenced by diverse factors such as company performance, macroeconomic conditions, policy changes, and global financial events.

Risk represents the possibility that realized returns may differ from what investors expect. It encompasses both downside potential (losses) and upside variability (unexpected gains). Thus, risk is not inherently negative; higher-risk investments may generate higher returns,

though they also carry a greater probability of loss. Understanding this dual nature of risk is essential for investors, as it forms the basis of the risk-return trade-off and guides the construction of efficient portfolios.

In equity markets, risk is broadly classified into **systematic risk** and **unsystematic risk**, which together determine the total risk of an investment.

1. Systematic Risk

Systematic risk refers to the risk that affects the entire market or a major segment of it. It arises due to external factors beyond the control of individual companies and cannot be eliminated through diversification. Since this type of risk influences all securities to some extent, investors have to bear it as an unavoidable part of investing in the stock market. Systematic risk is generally measured using beta, which indicates the sensitivity of a stock's returns in relation to movements in the overall market.

Sources of Systematic Risk

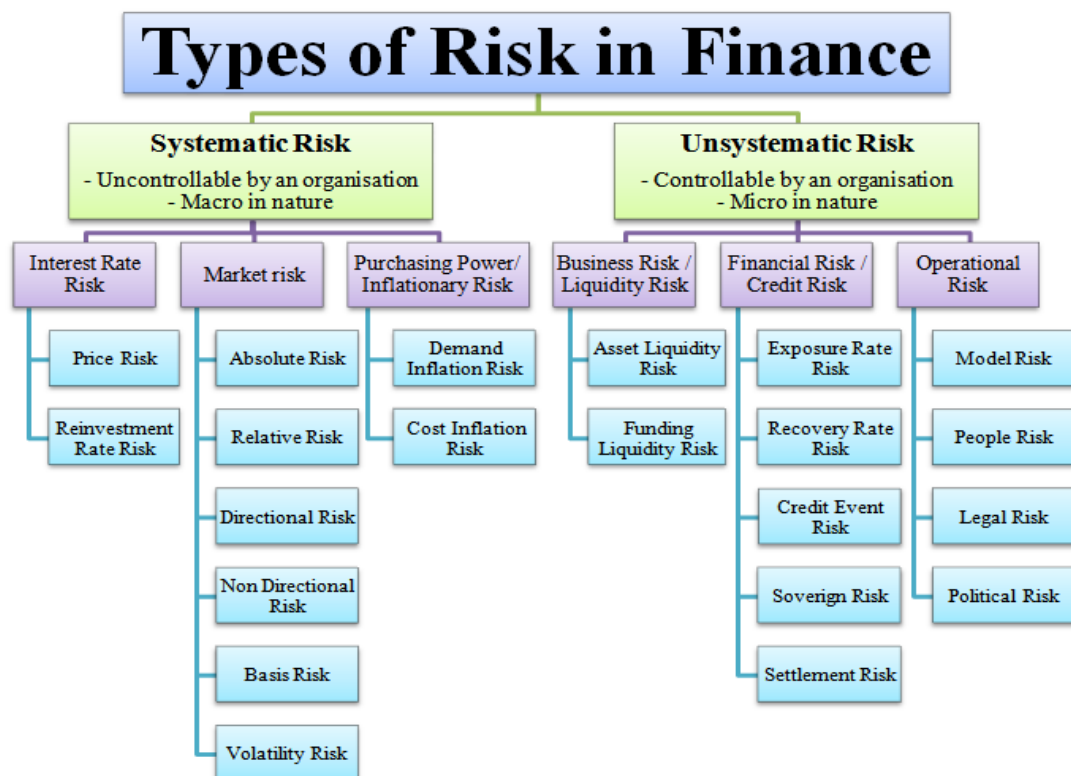
Systematic risk originates from broad economic and market-related factors. The major sources of systematic risk include changes in interest rates, fluctuations in inflation, economic conditions such as recession or expansion, political instability, government policy changes, exchange rate variations, and unexpected global events such as wars, natural disasters, or pandemics. These factors affect the overall market and, therefore, influence the performance of most investments.

2. Unsystematic Risk

Unsystematic risk refers to the risk that is specific to an individual company, industry, or sector. It arises from internal factors that directly affect the operations and performance of a particular organisation. Unlike systematic risk, unsystematic risk can be reduced or eliminated through diversification by investing in a variety of securities across different industries.

Sources of Unsystematic Risk

Unsystematic risk arises from company-specific factors that may adversely affect business performance. These factors include ineffective management decisions, poor financial performance, labour strikes, operational inefficiencies, product failures, technological issues, intense competition within the industry, and legal or regulatory problems faced by the company. Since these risks are limited to specific firms or sectors, they can be managed through proper diversification of investments.



I. Systematic Risk

1. Interest Rate Risk

Interest rate risk refers to the sensitivity of equity markets to changes in interest rates. When interest rates rise, borrowing costs increase, corporate profits decrease, and stock prices fall. When interest rates decline, borrowing becomes cheaper, leading to higher investment and improved stock valuations. This risk is more significant in sectors like banking, real estate, and capital-intensive industries.

- a. **Price Risk:** Rising discount rates reduce the present value of future cash flows, lowering stock prices.
- b. **Reinvestment Risk:** Future earnings may be reinvested at lower interest rates, reducing returns.

2. Market Risk:

Market risk refers to the risk of losses due to overall movements in the stock market caused by economic conditions, global events, or investor sentiment. It affects all securities and cannot be eliminated through diversification.

- a. **Absolute Risk:** Total potential loss in value.
- b. **Relative Risk:** Loss compared to benchmark indices.
- c. **Directional Risk:** Exposure to upward or downward market trends.
- d. **Volatility Risk:** Uncertainty due to rapid price swings.
- e. **Basis Risk:** Mismatch between related securities or derivatives.

3. Inflation Risk

Inflation risk refers to the decrease in real investment returns due to rising prices. It increases business costs and reduces consumer purchasing power, which can negatively affect corporate profits and stock performance.

- a. **Demand-Pull Inflation:** Price increase due to high demand.
- b. **Cost-Push Inflation:** Price increase due to rising input costs.

II. Unsystematic Risk

1. Business Risk

Business risk refers to the uncertainty arising from a company's internal operations and performance. It is influenced by factors such as management efficiency, production process, product demand, and competitive position in the market. Poor business decisions or declining sales can negatively impact a company's profitability and stock price.

- a. **Operational Risk:** Inefficiencies in day-to-day business activities.
- b. **Sales Risk:** Fluctuations in product demand and revenue.

2. Financial Risk

Financial risk stems from a company's capital structure and its ability to meet financial obligations. Firms with high debt levels are more exposed, as interest payments and repayment obligations can strain cash flows. Credit risk, recovery risk, and exposure to default are all part of financial risk. Equity investors are particularly sensitive to leverage ratios, since excessive borrowing increases vulnerability during downturns.

- a. **Leverage Risk:** Risk due to high debt financing.
- b. **Liquidity Risk:** Difficulty in meeting short-term financial obligations.

3. Operational Risk

Operational risk is linked to failures in internal processes, systems, or human resources. Errors in financial reporting, technology breakdowns, fraud, or poor governance can all damage investor confidence. Legal disputes and compliance failures also fall under operational risk. For equity investors, operational weaknesses can lead to sudden declines in stock prices even when market conditions are stable.

Measurement of Risk in Equity Markets

Risk in equity markets is commonly evaluated using statistical techniques that help in understanding the variability and uncertainty of returns. These measures form an important part of portfolio analysis and are widely used in financial decision-making.

There are five principal risk measures:

- **Standard Deviation:** It indicates how much the returns of a stock deviate from its average return. Greater deviation reflects higher uncertainty and risk.
- **Variance:** It represents the squared deviations of returns from the mean and helps in measuring the overall spread of data.
- **Beta (β):** It shows the responsiveness of a stock's returns to changes in the overall market. A beta greater than one suggests that the stock is more volatile than the market.
- **Value at Risk (VaR):** It estimates the possible maximum loss that an investment may face over a given time period at a specified confidence level.
- **Sharpe Ratio:** It evaluates the return earned relative to the total risk taken, helping investors understand whether higher returns justify the level of risk involved.

RETURN ANALYSIS IN EQUITY MARKETS

Investment means putting money into financial assets with the expectation of earning profit or increasing wealth in the future. In equity markets, investors mainly invest in shares to earn returns in the form of dividends and capital appreciation. However, the returns from equity investments are not fixed because they depend on market conditions, company performance, risk, inflation, and investment period.

Generally, investors expect higher returns when they invest for a longer duration and take higher levels of risk. Investors also prefer returns that are greater than the inflation rate so that the real value of their investment increases over time. Therefore, return analysis becomes important in evaluating the performance of investments in equity markets.

The expected return on an investment is affected by factors such as risk, inflation, and the time value of money. The nominal risk-free rate includes the effect of inflation, while the real risk-free rate does not consider inflation. Investors also expect an additional return called risk premium for taking extra risk in the market.

The nominal risk-free rate can be calculated using the following formula:

$$\text{NRFR} = (1 + \text{RRFR}) \times (1 + \text{Inflation Rate}) - 1$$

The expected return on investment is calculated as:

$$\text{Expected Return} = \text{NRFR} + \text{Risk Premium}$$

The return from an equity share includes dividend income and capital gain or loss. It can be calculated using the following formula:

$$R = \frac{[D_1 + (P_1 - P_0)]}{P_0}$$

Where:

- R = Rate of return
- D₁ = Dividend received
- P₀ = Initial price of the share
- P₁ = Final price of the share

Average return is used to measure the overall performance of an investment over different periods. It is calculated as:

$$\text{Average Return} = \frac{(R_1 + R_2 + \dots + R_n)}{n}$$

Expected return can also be calculated by assigning probabilities to different possible outcomes. This helps investors estimate future returns under different market situations.

$$E(R) = \sum (R_i \times P_i)$$

Where:

- E(R) = Expected return
- R_i = Return in each situation
- P_i = Probability of each return

Overall, return analysis helps investors understand the profitability of investments and supports them in selecting suitable investment opportunities based on expected return and market risk.

Types of Return in Equity Markets

1. Capital Gain Return

Capital gain arises when the selling price of a stock is higher than its purchase price. It reflects the appreciation in the value of the investment and is largely affected by market trends and the company's performance.

2. Dividend Return

Dividend return is the portion of profit distributed by a company to its shareholders. It acts as a source of periodic income and is generally observed in well-established companies with consistent earnings.

3. Total Return

Total return combines both price appreciation and dividend income earned from an investment. It gives a complete view of how much an investor gains from holding a stock.

4. Expected Return

Expected return refers to the estimated return that an investor anticipates earning based on past data, risk level, and market conditions. It is used as a basis for making investment decisions.

5. Realized Return

Realized return is the actual gain or loss obtained after the investment period ends or when the asset is sold. It represents the true outcome of the investment.

6. Annualized Return (CAGR)

Annualized return shows the average rate at which an investment grows each year over a period of time, taking into account the effect of compounding. It helps in comparing performance across different investments.

RISK AND RETURN TRADE-OFF

The relationship between risk and return is a key concept in investment decision-making. It indicates that the possibility of earning higher returns usually comes with an increase in the level of risk involved. Investors must therefore accept a certain degree of uncertainty in order to achieve better returns.

In equity markets, returns are not guaranteed and are influenced by factors such as price fluctuations, company performance, and overall economic conditions. Because of this uncertainty, investors expect additional returns as compensation for taking higher risk.

Low-risk investment options, such as government securities, generally provide stable but comparatively lower returns. On the other hand, equity investments have the potential to generate higher returns, but they are subject to greater volatility and market uncertainty.

The connection between risk and return is also supported by financial models like the Capital Asset Pricing Model (CAPM), which explains that expected return increases with the level of systematic risk, measured by beta.

The return on an investment can be calculated using the following formula:

$$\text{Return} = \frac{\text{Amount Received} - \text{Amount Invested}}{\text{Amount Invested}}$$

Overall, investors need to carefully balance risk and return based on their financial goals and risk tolerance while making investment decisions.

RISK-RETURN RELATIONSHIP

The risk-return relationship is a fundamental concept in finance which explains that the level of expected return is directly linked to the amount of risk taken by an investor. Generally, investments that carry higher uncertainty tend to offer higher potential returns, while safer investments usually provide lower returns. This is because investors require compensation, known as risk premium, for taking additional risk.

In equity markets, returns are not fixed and are influenced by various factors such as market fluctuations, economic conditions, inflation, interest rate changes, and company

performance. As a result, equities are considered riskier than fixed-income or risk-free investments like government securities, but they also offer greater return potential.

The relationship between risk and return is therefore positive in nature, and investors need to balance both based on their financial goals and risk tolerance. This concept is widely used in investment decision-making and portfolio construction, especially under models like the Capital Asset Pricing Model (CAPM), where expected return increases with higher systematic risk (beta).



In practical investment decisions:

Low-risk investments → low return

High-risk investments → high return potential

The risk–return trade-off helps investors make rational investment decisions by optimizing returns for a given level of risk.

2.2 Previous Study

International Studies

The concept of risk and return has been widely studied in financial economics as it forms the basis of investment decision-making. Researchers have developed various theories and models to explain how risk affects expected returns in financial markets. These frameworks are extensively used in portfolio analysis, asset pricing, and equity valuation, especially in investment institutions and brokerage firms such as Ventura Securities Ltd.

Markowitz Modern Portfolio Theory (1952): Harry Markowitz introduced Modern Portfolio Theory in 1952, which explains that investors can reduce overall risk by holding a diversified portfolio. According to his theory, risk is measured using variance or standard deviation, and investors aim to achieve the best possible return for a given level of risk. This concept highlights the importance of diversification in minimizing unsystematic risk.

Capital Asset Pricing Model (CAPM): The Capital Asset Pricing Model (CAPM), developed by Sharpe, Lintner, and Mossin, explains that expected return is linked to systematic risk, measured through beta. It states that investors are compensated only for market-related risk, assuming efficient markets and rational investors. CAPM is widely used in estimating expected returns and evaluating investment performance.

Efficient Market Hypothesis (Fama, 1970): Eugene Fama's Efficient Market Hypothesis suggests that stock prices reflect all available information, making it difficult to consistently earn abnormal returns. However, in developing markets, price inefficiencies may still exist due to information gaps and behavioral factors.

Fama and French Model (1992, 1993): Fama and French later expanded asset pricing theory by introducing additional risk factors such as firm size and value characteristics. Their findings show that returns are influenced by multiple variables, and not just market risk alone.

Empirical Studies in Emerging and Indian Markets

Studies conducted in the Indian market have tested the applicability of global financial models and found mixed results. Research indicates that beta alone may not fully explain stock returns in emerging economies.

Some studies such as **Gupta and Sehgal (1993)** observed that the Capital Asset Pricing Model does not fully hold in India, suggesting inefficiencies in the market. Similarly, **Basu (2004)** found that fundamental ratios like earnings-to-price significantly influence stock returns.

Further research by **Dhankar and Kumar (2007)** suggested that returns are affected by multiple economic and company-specific factors beyond systematic risk. **Raju and Ghosh (2011)** also found that volatility plays an important role in explaining returns, although beta alone is insufficient.

Gudimetla, Aruna. (2015)

The study was conducted to examine the market timing and selectivity abilities of fund managers managing Indian sector mutual funds. The performance of selected sector mutual funds was evaluated using established performance measurement models and risk-adjusted return indicators. The findings revealed varying levels of fund manager effectiveness in market timing and stock selection across different sector funds. The study concluded that selectivity skills contributed more significantly to fund performance than market timing abilities and highlighted the importance of managerial expertise in achieving superior investment returns.

Gudimetla, A. (2018)

The study aimed to examine whether thematic mutual fund schemes were able to generate satisfactory returns and maintain consistent performance over a period of time. Various risk-adjusted performance measures were used to assess and compare the returns of selected thematic funds with their benchmark indices. The findings indicated that while some thematic funds outperformed the market during certain periods, their superior performance was not consistently maintained. The study concluded that past performance could not be considered

a reliable predictor of future returns and provided valuable insights for investors regarding thematic mutual fund investments.

Gudimetla, A., & Raghunatha Reddy, D. (2018)

The study aimed to analyse whether sector-specific mutual fund schemes were capable of generating superior returns in relation to the risks assumed by investors. The performance of selected sectoral mutual funds was assessed using various risk-adjusted performance measures and benchmark comparisons. The study concluded that sector-specific mutual funds could offer attractive investment opportunities; however, investors should carefully consider sectoral risks and market trends before making investment decisions.

Kumar and Choudhary (2021)

Kumar and Choudhary (2021), in their study titled *"Risk & Return Analysis: Evidence from the Indian Equity Market,"* examined the relationship between risk and return using selected equities from the Indian stock market. The study employed statistical tools such as standard deviation to measure risk and analysed returns based on stock price movements. The findings indicated that although higher risk is generally associated with higher returns, this relationship is not uniform across all securities. The study emphasized the significance of diversification in minimizing unsystematic risk and highlighted the importance of considering risk-adjusted returns while making investment decisions.

Schoenmaker and Schramade (2023)

Schoenmaker and Schramade (2023), in their chapter *"Risk-Return Analysis"* from *Corporate Finance for Long-Term Value*, examined the relationship between risk and return from a long-term value perspective. The authors emphasized that investment decisions should consider both expected returns and the associated risks to ensure sustainable value creation. The study discussed various tools such as standard deviation, beta, diversification, and risk-adjusted return measures for evaluating investment performance. It also highlighted the importance of incorporating environmental, social, and governance (ESG) factors into financial decision-making. The authors concluded that effective risk-return analysis supports informed investment decisions, efficient resource allocation, and long-term financial stability.

CHAPTER – III
INDUSTRY PROFILE
AND
COMPANY PROFILE

3. INDUSTRY PROFILE

3.1 INTRODUCTION

Financial Services Industry in India

The financial services sector in India plays a vital role in supporting economic development by channeling savings into productive investment opportunities. It includes key segments such as banking, insurance, mutual funds, capital markets, and stock broking services. Among these, the capital market is especially important as it enables companies to raise funds and provides investors with opportunities to generate returns through equity investments.

The Indian financial system operates under a well-defined regulatory framework designed to ensure transparency and safeguard investor interests. The Securities and Exchange Board of India is the principal authority responsible for regulating capital markets and overseeing stock broking activities. It establishes guidelines to maintain fair trading practices, reduce fraudulent activities, and build investor confidence.

Over time, the financial services industry in India has undergone significant transformation due to technological progress and digitalization. The introduction of online trading platforms and mobile applications has made financial markets more accessible, encouraging greater participation from retail investors. Additionally, initiatives promoting financial inclusion have further expanded the reach of investment services across different sections of society.

Overview of the Indian Capital Market Industry

The Indian capital market is considered one of the fastest-evolving segments of the financial system, contributing significantly to economic growth and investment activity. It is broadly classified into two main components: the primary market and the secondary market. The primary market allows companies to mobilize funds directly from investors by issuing new securities through instruments such as initial public offerings, rights issues, and follow-on offerings. In contrast, the secondary market provides a platform for trading existing

securities, ensuring liquidity and enabling investors to enter or exit investments conveniently.

The functioning of the capital market in India is supported by major stock exchanges like the National Stock Exchange of India and Bombay Stock Exchange. These exchanges offer an organized and transparent system for buying and selling securities, which helps in accurate price determination and builds trust among investors.

The expansion of the Indian capital market can be attributed to several factors, including economic reforms, advancements in technology, improved corporate governance, and rising investor awareness. Increased participation from both domestic and international investors has further strengthened market depth and efficiency, making it an important avenue for investment and capital formation.

Stock Broking Industry in India

The stock broking industry plays a vital role in the functioning of the capital market by acting as a link between investors and stock exchanges. Brokers facilitate the buying and selling of securities on behalf of clients and provide access to various financial instruments such as equities, derivatives, mutual funds, and initial public offerings. Along with trade execution, many brokerage firms also offer additional services including investment research, portfolio management, financial advisory, and risk assessment.

In recent times, the industry has experienced rapid changes due to advancements in technology and evolving investor preferences. The introduction of digital trading platforms, mobile-based applications, and real-time market information has simplified the investment process and improved efficiency. The presence of both full-service brokers and discount brokers has intensified competition, resulting in lower transaction costs and better service offerings.

The industry operates under the supervision of the Securities and Exchange Board of India, which ensures fair practices, transparency, and protection of investor interests. These regulatory measures have contributed to building trust and stability within the stock broking sector, making it more organized and dependable for market participants.

Role of Stock Brokers in Equity Markets

Stock brokers perform an important function in equity markets by serving as intermediaries between investors and stock exchanges. They enable investors to participate in buying and selling securities by executing trades efficiently and providing access to market platforms. Brokers also supply relevant market data, analysis, and research inputs that assist investors in understanding market conditions.

Apart from facilitating transactions, brokers offer guidance and advisory support to help investors take informed decisions. Their recommendations are often based on factors such as market trends, company fundamentals, and overall economic conditions. This advisory role helps investors align their investment choices with their financial goals.

In addition, stock brokers support portfolio management by encouraging diversification across various sectors and financial instruments, which helps in reducing investment risk. They also play a role in improving financial awareness by educating investors about market behavior, trading techniques, and long-term investment planning. This contribution enhances investor confidence and promotes more disciplined participation in the equity market.

Growth and Development of the Broking Industry

The stock broking industry in India has witnessed significant expansion in recent years, driven by both technological progress and structural changes in the financial sector. The increasing availability of internet connectivity and smartphones has made online trading platforms easily accessible, encouraging a rise in retail investor participation. At the same time, improved financial awareness has motivated individuals to explore investment opportunities in equities and mutual funds.

Another key driver of growth is the decline in brokerage fees along with the emergence of low-cost and discount brokerage models. These changes have made trading more affordable and attractive to a wider range of investors. Technological innovations such as algorithmic trading, artificial intelligence, and advanced data analytics have further enhanced trading efficiency and decision-making capabilities.

Moreover, the involvement of institutional participants, including mutual funds, foreign investors, and insurance companies, has contributed to greater market liquidity and stability. As the market becomes more sophisticated, there is an increasing need for analytical approaches, particularly risk and return analysis, to support informed investment decisions.

Importance of Risk and Return Analysis in the Broking Industry

Risk and return analysis is an essential component of the stock broking industry, as it supports both investors and financial advisors in making well-informed investment decisions. It helps in assessing the level of uncertainty or volatility associated with different securities and determining whether the potential returns justify the level of risk involved. This evaluation enables investors to choose investments that match their financial goals and risk-bearing capacity.

Brokerage firms apply various quantitative tools such as beta, standard deviation, variance, and the Sharpe ratio to evaluate risk-adjusted performance. These measures assist in comparing multiple investment alternatives and identifying securities that offer an appropriate balance between risk and return. By using such analytical techniques, brokers can provide more reliable and data-driven recommendations to their clients.

In firms like Ventura Securities Ltd, risk and return analysis serves as a core element of research and advisory services. It plays a crucial role in portfolio construction, helping investors achieve diversification and optimize returns while controlling overall risk exposure.

Role of Ventura Securities Ltd in the Industry

Ventura Securities Ltd operates as a full-service brokerage firm in India, offering a comprehensive range of financial services such as equity and derivatives trading, mutual fund investments, initial public offerings, portfolio management services, and equity research. The firm plays a significant role in assisting investors by delivering research-oriented insights and analytical support that aid in evaluating investment opportunities.

The company emphasizes structured investment planning through its advisory services and model-based portfolio strategies. It incorporates financial frameworks like the Capital Asset Pricing Model, along with performance measures such as the Sharpe Ratio and Treynor

Ratio, to assess risk-adjusted returns and overall portfolio efficiency. By combining quantitative analysis with market research, the firm supports investors in identifying suitable securities and building diversified investment portfolios aligned with their financial objectives.

The stock broking industry in India continues to evolve rapidly due to advancements in technology, strong regulatory oversight, and growing participation from retail and institutional investors. Within this dynamic environment, the evaluation of risk and return remains a fundamental aspect of investment decision-making. Firms like Ventura Securities Ltd contribute to this ecosystem by offering advisory services, research-based recommendations, and analytical tools that enable investors to make informed choices.

In this context, risk and return analysis serves as a critical foundation for developing effective investment strategies. It helps investors balance potential returns with associated risks, thereby supporting long-term portfolio growth and financial stability in the Indian capital market.

COMPANY PROFILE

3.2 INTRODUCTION

Ventura Securities Ltd is a recognized full-service brokerage and financial services firm in India, offering a wide range of investment and wealth management solutions to retail investors, high-net-worth individuals, and institutional clients. The company operates within the Indian capital market framework, which is regulated by the Securities and Exchange Board of India to ensure transparency, investor protection, and fair market practices. It acts as an intermediary by facilitating investor participation in stock exchanges such as the National Stock Exchange of India and Bombay Stock Exchange.

The firm provides a diverse set of services, including equity and derivatives trading, currency trading, mutual fund investments, initial public offerings, and portfolio management services. In addition to trade execution, the company places strong emphasis on research and advisory support by offering market analysis, stock recommendations, and insights that assist investors in understanding market behavior and making informed financial decisions.

To remain competitive in the evolving brokerage industry, Ventura Securities Ltd has integrated advanced technology into its operations. It offers digital trading platforms, mobile-based applications, and real-time tracking systems that enhance user experience and improve access to financial markets, particularly for retail participants.

The company also highlights the importance of evaluating investments through risk and return analysis. It promotes the use of financial tools such as beta, standard deviation, variance, the Sharpe ratio, and the Capital Asset Pricing Model to assess volatility and expected returns. These techniques support investors in constructing diversified portfolios aligned with their risk preferences and financial objectives.

Over time, Ventura Securities Ltd has established a strong presence in the financial services sector by focusing on transparency, client-oriented services, and research-driven strategies. Its efforts toward investor awareness and structured investment planning contribute to the overall development and efficiency of the Indian capital market.

Overview of the Company

Ventura Securities Ltd operates as a financial services provider in India, primarily engaged in facilitating investment and trading activities in the capital market. The firm caters to a diverse client base, including individual investors, high-net-worth clients, and institutional participants. Its operations function within the regulatory environment governed by the Securities and Exchange Board of India, which promotes fair practices and safeguards investor interests.

The company enables participation in multiple market segments such as equities, derivatives, mutual funds, and public issues. In addition to executing transactions, it offers analytical support and market insights to assist clients in evaluating investment choices.

By utilizing digital trading systems and emphasizing risk–return assessment, Ventura Securities Ltd facilitates informed decision-making and supports investors in managing their portfolios effectively.

3.2.1 VISION, MISSION & VALUES

Vision

The vision of Ventura Securities Ltd is to establish itself as a prominent financial services provider in India by delivering innovative, technology-enabled, and customer-oriented investment solutions. The company aspires to support investors in building sustainable wealth and achieving long-term financial security.

Mission

The mission of Ventura Securities Ltd is to offer reliable and efficient financial services through transparent trading systems and advanced technological platforms. It focuses on providing research-driven insights and guidance that enable investors to make well-informed decisions. The company also aims to promote financial awareness and simplify the investment process, helping clients achieve favorable risk-adjusted returns in the capital market.

Values

Ventura Securities Ltd is guided by a set of core values that shape its operations and client relationships. The organization emphasizes integrity by ensuring ethical practices and transparency in all activities. It maintains a strong customer-centric approach by prioritizing client needs and delivering consistent value. Innovation is encouraged through continuous improvement in financial services and technology. The company is committed to excellence in research and service delivery, while also promoting responsible investing practices. Additionally, it ensures trust and compliance by adhering strictly to regulatory standards and fostering long-term client relationships.

3.2.2 BUSINESS PHILOSOPHY

The business philosophy of Ventura Securities Ltd is based on the idea that wealth creation in capital markets requires a systematic and disciplined approach, with equal emphasis on both risk management and return generation. The organization believes that effective investment outcomes are achieved through informed decision-making, supported by strong research and structured financial analysis.

The company follows a client-focused approach by understanding the unique financial goals and risk preferences of its investors and offering tailored investment solutions. It aims to make financial markets more transparent, accessible, and efficient for a wide range of clients, including retail investors, high-net-worth individuals, and institutional participants.

A significant element of its philosophy is the reliance on research-based investment strategies. The firm emphasizes the importance of equity research, market evaluation, and data-driven insights to guide investment decisions, minimizing dependence on speculative practices.

In addition, Ventura Securities Ltd promotes a balanced investment approach by encouraging diversification and disciplined portfolio management. This helps investors manage market fluctuations while working toward consistent long-term returns.

The organization also integrates advanced technological tools into its operations to enhance trading efficiency, improve accessibility, and support better decision-making. Overall, the

company is committed to fostering long-term client relationships by promoting responsible investing and sustainable wealth creation.

Strengths of the Company

Ventura Securities Ltd possesses several key strengths that support its position in the financial services industry:

- **Research-Based Guidance:** The company provides detailed market analysis and investment insights, helping clients take well-informed decisions.
- **Advanced Trading Technology:** It offers digital trading platforms with real-time data and user-friendly features, enabling efficient execution of trades.
- **Comprehensive Service Offering:** A variety of financial services are available, allowing clients to access multiple investment options under one platform.
- **Client-Centric Approach:** The firm focuses on understanding investor needs and delivering solutions that align with their financial goals.
- **Focus on Risk–Return Evaluation:** It emphasizes analyzing both potential returns and associated risks, supporting balanced and informed investment strategies.

MAJOR USERS

Ventura Securities Ltd serves a diverse group of investors and market participants who utilize its trading, brokerage, and advisory services for investment and wealth management purposes. The major categories of users include:

- Retail Investors
- High Net-Worth Individuals (HNIs)
- Institutional Investors
- Active Traders
- Financial Advisors and Intermediaries

All these user groups depend on the services of Ventura Securities Ltd to effectively participate in capital markets while managing risk and optimizing returns.

3.2.3 WORK CULTURE

Dynamic and Performance-Oriented Culture

The work culture at Ventura Securities Ltd is fast-paced, professional, and strongly oriented toward performance. Employees are encouraged to remain proactive, flexible, and responsive to evolving market trends and client expectations. Individual performance is assessed based on efficiency, output, and contribution toward organizational objectives.

Learning and Development Orientation

The organization promotes continuous learning by providing employees with exposure to financial markets, investment instruments, and analytical techniques. This environment supports the development of critical thinking and enhances decision-making skills, which are essential in areas such as equity research and risk-return evaluation.

Team Collaboration and Coordination

A culture of teamwork is maintained across departments including research, trading, advisory, operations, and compliance. Effective coordination among these functions ensures seamless service delivery and supports the creation of well-structured investment solutions.

Technology-Driven Workplace

The company adopts modern trading platforms and digital tools to improve operational efficiency. Real-time data systems and analytical applications assist employees in making accurate and timely decisions, thereby enhancing the quality of services provided to clients.

Ethics and Compliance

Maintaining ethical practices and regulatory compliance is a key aspect of the organizational culture. Employees are expected to follow the guidelines set by the Securities and Exchange Board of India and uphold transparency and integrity in all professional activities. This approach helps in building trust and maintaining long-term client relationships.

Professional Growth and Accountability

The company encourages employees to take responsibility for their roles while providing opportunities for professional advancement. A performance-driven environment supports career growth and motivates individuals to maintain high standards of work quality.

3.2.4 HR CONCEPTS

Ventura Securities Ltd acknowledges that organizational success in the financial services sector is largely driven by the competence, motivation, and professionalism of its employees. The company's human resource approach focuses on talent development, employee involvement, performance orientation, and adherence to ethical standards.

Talent Acquisition & Development

The organization recruits individuals with strong expertise in finance, research, and client advisory services. Newly hired employees undergo structured orientation programs that introduce them to equity markets, regulatory requirements, and trading platforms. Ongoing training initiatives ensure that employees stay updated with market developments, Securities and Exchange Board of India guidelines, and risk–return evaluation techniques.

Performance Management

The company follows a performance-driven system where employees are assessed based on measurable indicators such as client satisfaction, quality of advisory services, and compliance with regulations. Performance evaluations are linked to rewards and incentives, ensuring that individual efforts are aligned with organizational objectives.

Employee Engagement & Motivation

Ventura Securities Ltd promotes a supportive and engaging work environment through activities such as training workshops, team interactions, and recognition programs. Incentive mechanisms are designed to reward both individual performance and team contributions, encouraging consistent productivity.

Learning & Knowledge Sharing

The organization encourages continuous learning by supporting participation in professional certifications, seminars, and internal research discussions. Knowledge-sharing practices enable employees to exchange ideas and insights, enhancing overall expertise in financial analysis and investment decision-making.

Ethics & Compliance Orientation

Maintaining ethical conduct is a key priority within the company. Employees are trained to follow transparent practices and comply with regulatory standards set by authorities like Securities and Exchange Board of India. This ensures responsible service delivery and strengthens client trust.

3.2.5 HIERARCHY (BOARD OF DIRECTORS)

Ventura Securities Ltd follows a well-defined organizational structure that supports effective governance, strategic planning, and efficient execution of business operations. The hierarchy ensures a clear distribution of responsibilities from top management to operational levels.

Board of Directors: The Board of Directors represents the highest decision-making authority within the organization. It is responsible for defining the company's vision, strategic direction, and long-term objectives. The Board also ensures adherence to regulatory requirements, including guidelines set by the Securities and Exchange Board of India, along with maintaining strong corporate governance practices.

Managing Director / Chief Executive Officer (CEO): The Managing Director or CEO is responsible for overseeing daily operations and implementing the policies formulated by the Board. This role ensures that the organization's strategies are effectively executed while maintaining a balance between risk management and return generation across all business activities.

Executive Directors: Executive Directors manage key functional areas such as equity research, trading operations, advisory services, and portfolio management. They are responsible for ensuring that each department operates efficiently and aligns with the overall business objectives of the organization.

Research & Advisory Division: This division is led by experienced analysts who conduct detailed market studies and prepare research reports. It focuses on evaluating investment opportunities, performing risk–return analysis, and providing recommendations to support informed decision-making.

Client Relationship Managers: Client Relationship Managers act as a link between the company and its clients. They provide personalized investment guidance, assist in portfolio construction, and ensure that investment strategies are aligned with the client’s financial goals and risk preferences.

Support Functions

- **Compliance and Legal Team:** Responsible for ensuring that all operations comply with regulatory frameworks and legal requirements, particularly those set by the Securities and Exchange Board of India.
- **Human Resources (HR):** Handles recruitment, employee development, and engagement initiatives.
- **Technology and Operations:** Manages trading platforms, digital systems, and back-office processes to ensure smooth functioning of services.

BOARD OF DIRECTORS

SAJID SIRAJ MALIK → DIRECTOR & CO-FOUNDER

HEMANT MAJETHIA → DIRECTOR, CEO & CO-FOUNDER

GANAPATHY VISHWANATHAN → DIRECTOR

MANU MONGA → DIRECTOR

JUZER YUSUF GABAJIWALA → DIRECTOR & COMPANY SECRETARY

SRIRAM SANKARANARAYANAN → DIRECTOR

CHAPTER – IV
DATA ANALYSIS
&
INTERPRETATION

ANALYSIS OF RETURN

4.1 Analysis of Return:

Analysis of Return refers to the process of measuring and evaluating the profit or loss generated from an investment over a period of time. It helps in understanding how much return an investor earns from a particular stock based on changes in its price.

$$\text{Rate of Return} = \frac{\text{Share price in the closing} - \text{Share price at the opening}}{\text{Share Price at the opening}}$$

$$\text{Rate of Return} = \frac{(P_1 - P_0)}{P_0}$$

P_0 = Opening (Current Price)

P_1 = Closing (Future Price)

4.2 Analysis of Returns of Selected Companies

The below tables explain the average return earned by the selected companies during the study period. It helps in understanding the overall return performance of the selected equity shares.

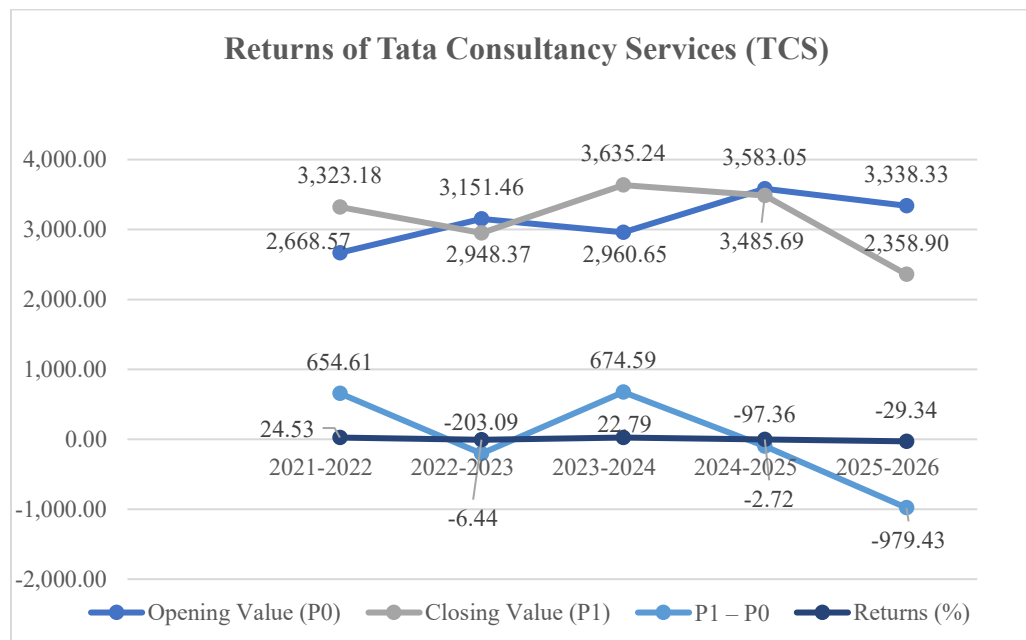
4.2.1. Calculation of Returns of Tata Consultancy Services (TCS)

Year-wise Return Calculation

Year	Opening Value (P ₀)	Closing Value (P ₁)	P ₁ - P ₀	Return $\frac{(P_1 - P_0)}{P_0} * 100$
2021-2022	2,668.57	3,323.18	654.61	24.53
2022-2023	3,151.46	2,948.37	-203.09	-6.44
2023-2024	2,960.65	3,635.24	674.59	22.79
2024-2025	3,583.05	3,485.69	-97.36	-2.72
2025-2026	3,338.33	2,358.90	-979.43	-29.34
Total Return				8.82

$$\text{Average Return} = \frac{8.82}{5} = 1.76$$

Return Analysis of Tata Consultancy Services (TCS)



Interpretation

The return analysis of Tata Consultancy Services indicates that the company experienced noticeable fluctuations in its performance during the study period. Although the stock generated positive returns in certain years, it also recorded negative returns, particularly a significant decline in the final year, which affected the overall performance. The average return of 1.76% reflects relatively low profitability and suggests that the stock did not provide consistent returns to investors. This pattern indicates moderate risk with limited return potential during the study period.

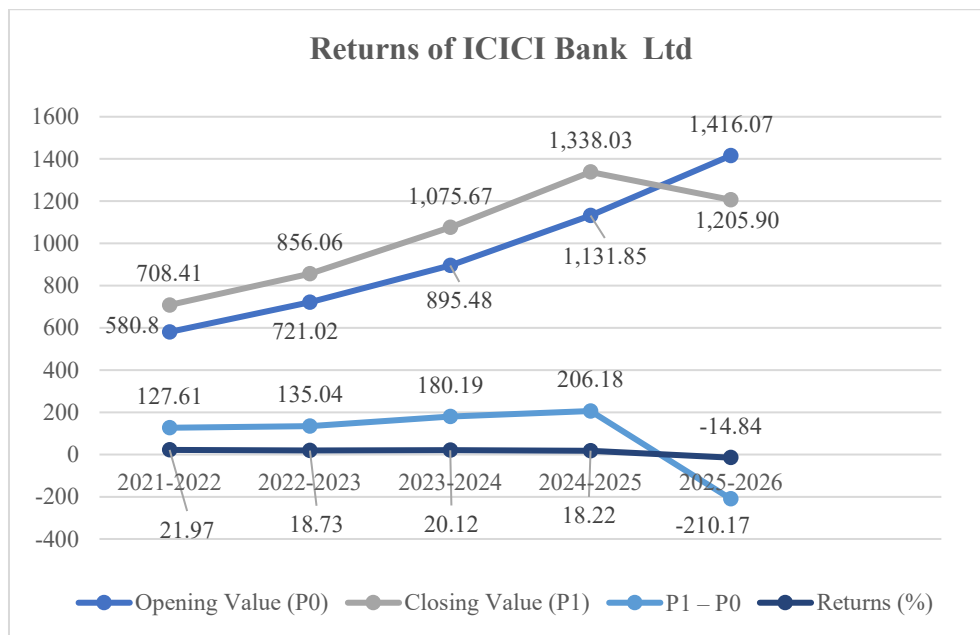
4.2.2. Calculation of Returns of ICICI Bank Limited

Year-wise Return Calculation

Year	Opening Value (P ₀)	Closing Value (P ₁)	P ₁ – P ₀	Return $\frac{(P_1 - P_0)}{P_0} * 100$
2021-2022	580.80	708.41	127.61	21.97
2022-2023	721.02	856.06	135.04	18.73
2023-2024	895.48	1,075.67	180.19	20.12
2024-2025	1,131.85	1,338.03	206.18	18.22
2025-2026	1,416.07	1,205.90	-210.17	-14.84
Total Return				64.20

$$\text{Average Return} = \frac{64.20}{5} = 12.84$$

Return Analysis of ICICI Bank Limited



Interpretation

The return analysis of ICICI Bank shows a strong and consistent upward performance for most of the study period. The stock delivered stable positive returns across multiple years, indicating steady growth and investor confidence. Although there was a negative return in the final year, it did not significantly impact the overall performance. With the highest average return of 12.84% among the selected stocks, ICICI Bank demonstrates strong profitability and comparatively lower volatility, making it an attractive option for investors seeking better returns.

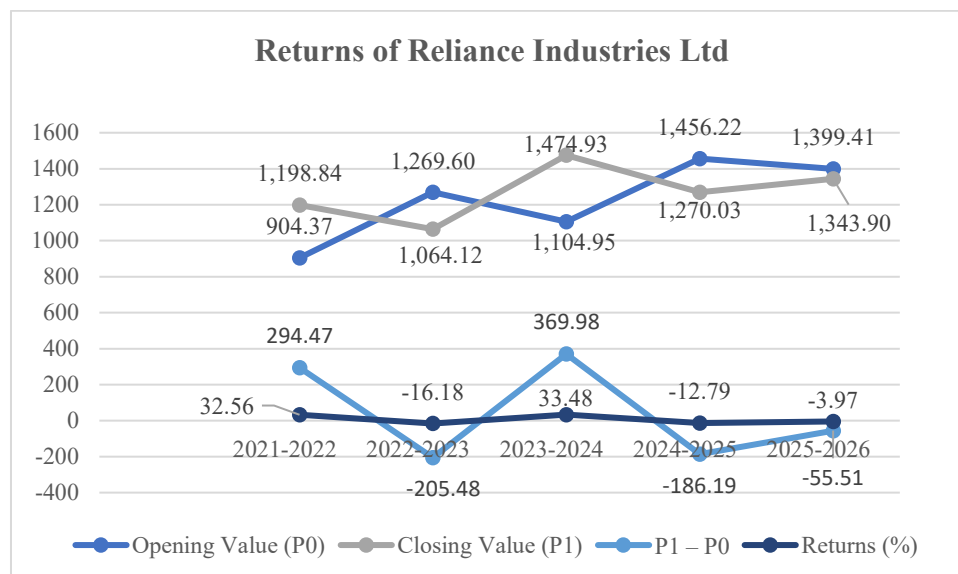
4.2.3. Calculation of Returns of Reliance Industries Limited

Year-wise Return Calculation

Year	Opening Value (P ₀)	Closing Value (P ₁)	P ₁ – P ₀	Return $\frac{(P_1 - P_0)}{P_0} * 100$
2021-2022	904.37	1,198.84	294.47	32.56
2022-2023	1,269.60	1,064.12	-205.48	-16.18
2023-2024	1,104.95	1,474.93	369.98	33.48
2024-2025	1,456.22	1,270.03	-186.19	-12.79
2025-2026	1,399.41	1,343.90	-55.51	-3.97
Total Return				33.11

$$\text{Average Return} = \frac{33.11}{5} = 6.62$$

Return Analysis of Reliance



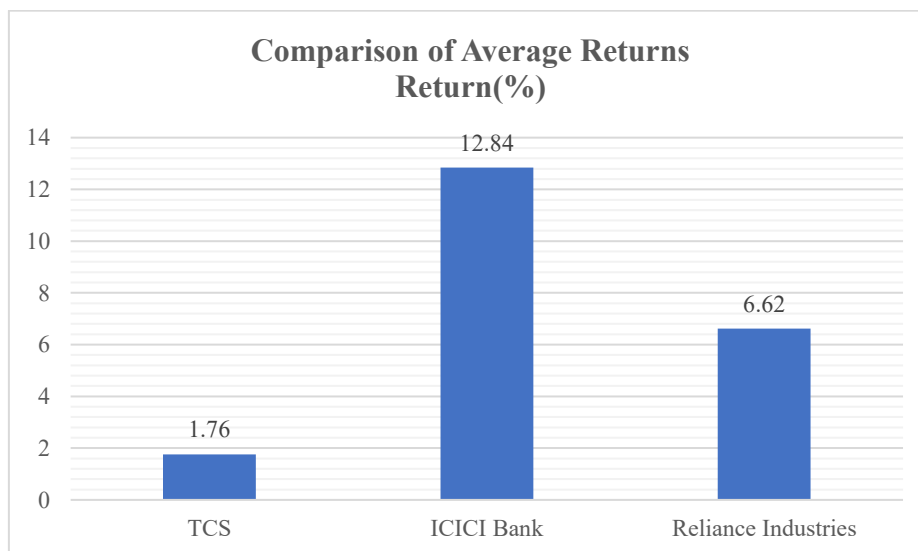
Interpretation

The return analysis of Reliance Industries reveals that the stock exhibited considerable fluctuations throughout the study period. The company recorded both high positive returns and notable negative returns, indicating variability in its performance. The average return of 6.62% suggests moderate profitability; however, the inconsistency in yearly returns

highlights a higher level of risk. This indicates that while the stock has the potential to generate good returns, it is also subject to market volatility and external factors.

4.2.4. Comparative Analysis of Average Returns of Selected Companies

Company	Average Return(%)
TCS	1.76
ICICI Bank	12.84
Reliance Industries	6.62



Interpretation

The return analysis indicates that ICICI Bank generated the highest average return among the selected companies during the study period. Reliance Industries recorded moderate returns, while TCS generated comparatively lower returns due to fluctuations in the IT sector and changing market conditions.

ANALYSIS OF RISK

Analysis of Risk

Analysis of risk is the measurement of fluctuations in investment returns, which indicates the level of uncertainty and volatility associated with a stock.

Standard Deviation

Standard deviation measures the degree of variation in stock returns from the average return, indicating the level of risk or volatility.

Formulas:

$$\text{Variance} \rightarrow \sigma^2 = \frac{\sum (R - \bar{R})^2}{n}$$

Where:

- R = Individual Term
- $\bar{R}$ = Average Return
- n = Number of years

$$\text{Standard Deviation} \rightarrow \sigma = \sqrt{\text{Variance}}$$

4.3 Risk Analysis using Standard Deviation

The following tables shows the risk level of selected stocks using standard deviation. It helps in identifying the fluctuations in stock returns during the study period.

4.3.1. Standard Deviation of Tata Consultancy Services (TCS)

Risk Analysis of TCS using Standard Deviation

Year	Return (R)	Average Return ($\bar{R}$)	Deviations ($R - \bar{R}$)	($R - \bar{R}$) ²
2021-2022	24.53	1.76	22.77	518.47
2022-2023	-6.44	1.76	-8.2	67.24
2023-2024	22.79	1.76	21.03	442.26
2024-2025	-2.72	1.76	-4.48	20.07
2025-2026	-29.34	1.76	-31.1	967.21
Total				2,015.25

$$\begin{aligned}\text{Variance} \rightarrow \sigma^2 &= \frac{\sum (R - \bar{R})^2}{n} \\ &= \frac{2015.25}{5} = 403.05\end{aligned}$$

$$\begin{aligned}\text{Standard Deviation} \rightarrow \sqrt{\text{Variance}} \\ &= \sqrt{403.05} = \mathbf{20.07}\end{aligned}$$

Interpretation

The analysis shows that Tata Consultancy Services (TCS) has an average return of 1.76%, but with a high standard deviation of 20.07%, indicating significant volatility in its yearly returns. This means the investment carries high risk, as returns fluctuate widely between strong gains and steep losses, making it suitable only for investors with a higher risk appetite while conservative investors may find it too unstable.

4.3.2. Standard Deviation of ICICI Bank Limited

Risk Analysis of ICICI using Standard Deviation

Year	Return (R)	Average Return ($\bar{R}$)	Deviations ($R - \bar{R}$)	($R - \bar{R}$) ²
2021-2022	21.97	12.84	9.13	83.36
2022-2023	18.73	12.84	5.89	34.69
2023-2024	20.12	12.84	7.28	53.00
2024-2025	18.22	12.84	5.38	28.94
2025-2026	-14.84	12.84	-27.68	766.18
Total				966.17

$$\begin{aligned}\text{Variance} \rightarrow \sigma^2 &= \frac{\sum (R - \bar{R})^2}{n} \\ &= \frac{966.17}{5} = 193.23\end{aligned}$$

$$\begin{aligned}\text{Standard Deviation} &\rightarrow \sqrt{\text{Variance}} \\ &= \sqrt{193.23} = \mathbf{13.9}\end{aligned}$$

Interpretation

The analysis of ICICI Bank shows that the average return is 12.84%, with a standard deviation of 13.9%, which indicates moderate volatility compared to TCS. This means that while ICICI has delivered relatively stable positive returns in most years, the sharp decline in 2025-26 (-14.84%) increased the overall risk. The lower standard deviation compared to TCS suggests that ICICI is a less risky investment, offering more consistent returns, though it is still exposed to market fluctuations. Overall, ICICI presents a balanced risk-return profile, making it more suitable for investors seeking steady growth with moderate risk.

4.3.3. Standard Deviation of Reliance Industries Limited

Risk Analysis of RELIANCE using Standard Deviation

Year	Return (R)	Average Return ($\bar{R}$)	Deviations ($R - \bar{R}$)	($R - \bar{R}$) ²
2021-2022	32.56	6.62	25.94	672.88
2022-2023	-16.18	6.62	-22.8	519.84
2023-2024	33.48	6.62	26.86	721.46
2024-2025	-12.79	6.62	-19.41	376.75
2025-2026	-3.97	6.62	-10.59	112.15
Total				2,403.08

$$\begin{aligned}\text{Variance} \rightarrow \sigma^2 &= \frac{\sum (R - \bar{R})^2}{n} \\ &= \frac{2403.08}{5} = 480.61\end{aligned}$$

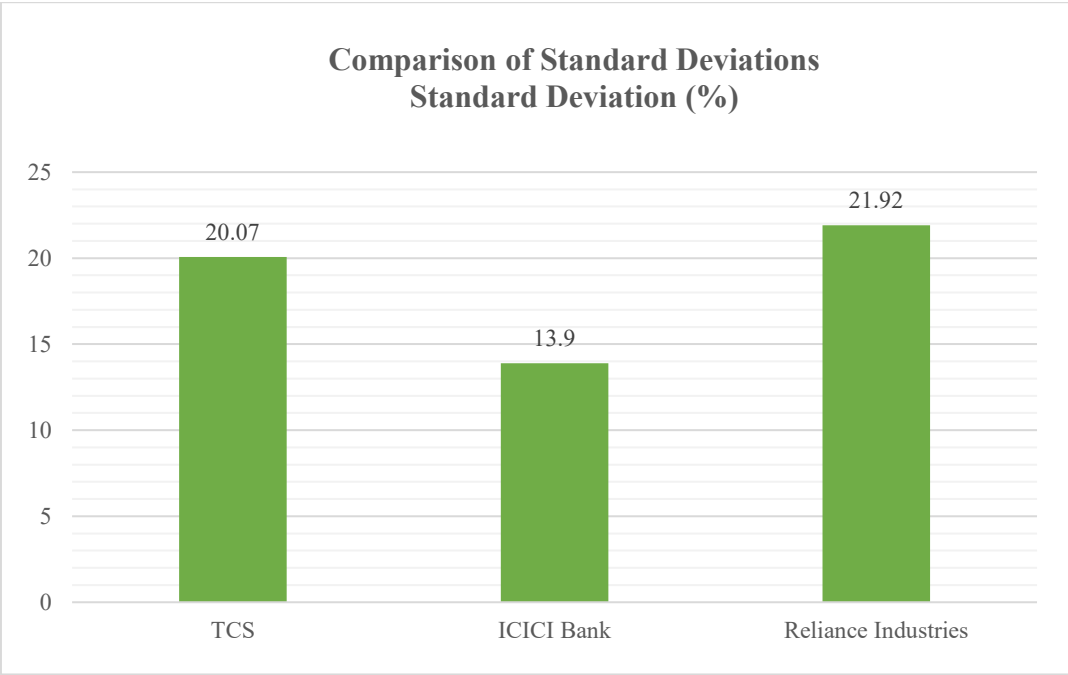
$$\begin{aligned}\text{Standard Deviation} &\rightarrow \sqrt{\text{Variance}} \\ &= \sqrt{480.61} = \mathbf{21.92}\end{aligned}$$

Interpretation

The analysis of Reliance Industries shows that the average return is 6.62%, but the standard deviation is 21.92%, which reflects very high volatility in its yearly returns. This means Reliance carries substantial risk, as returns fluctuate sharply between strong gains (above 30%) and steep losses (around -16% and -12%). The high standard deviation indicates that Reliance is a high-risk investment, with unpredictable performance, making it more suitable for aggressive investors seeking high potential returns, while conservative investors may find it too unstable for their portfolios.

4.3.4 Comparative Analysis of Standard Deviation of Selected Companies

Company	Standard Deviation (%)	Risk Level
TCS	20.07	High
ICICI Bank	13.90	Moderate
Reliance Industries	21.92	High

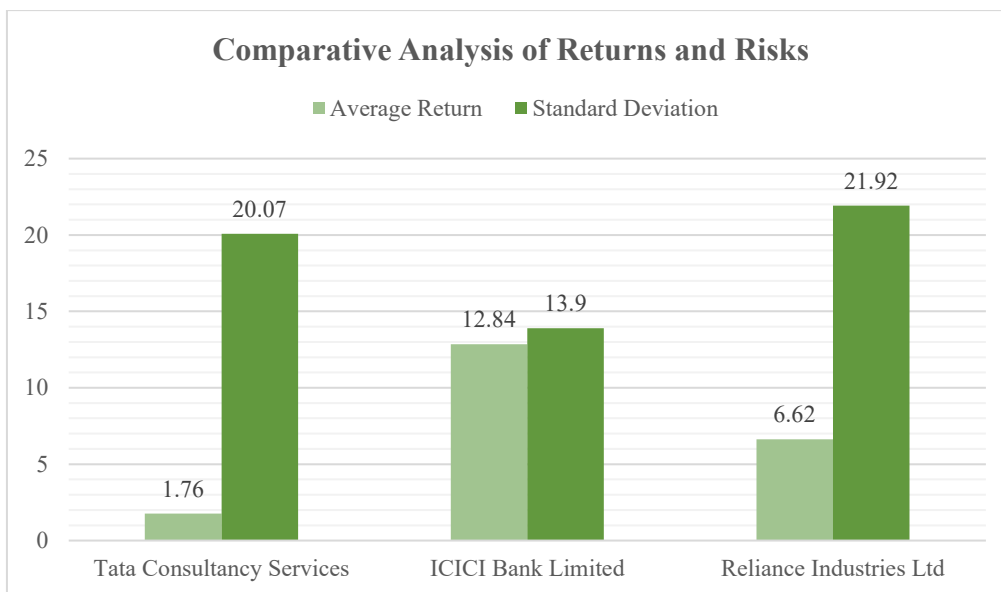


Interpretation

The standard deviation analysis shows that Reliance Industries carried the highest risk due to greater stock price fluctuations. ICICI Bank demonstrated comparatively lower volatility, indicating stable market performance during the study period.

4.3.5 Comparative Analysis of Average Return and Risks of Selected Companies

Company	Average Return	Standard Deviation
Tata Consultancy Services	1.76	20.07
ICICI Bank Limited	12.84	13.9
Reliance Industries Ltd	6.62	21.92



Interpretation

The comparison highlights that TCS delivers very low returns with high risk, showing poor stability for investors. ICICI Bank stands out with the highest average return and the lowest risk, making it the most reliable and consistent performer. Reliance offers moderate returns but carries the highest volatility, reflecting unpredictable performance. In summary, ICICI is the most suitable for investors seeking steady growth, while TCS and Reliance are more volatile options that may appeal to aggressive investors willing to accept greater uncertainty.

CALCULATION OF COVARIANCES

COVARIANCE

Covariance is a statistical measure that shows the relationship between the returns of two stocks. It indicates whether the returns move together or in opposite directions.

Formula

$$\text{COV}(X, Y) = \frac{\sum[(X - \bar{X})(Y - \bar{Y})]}{n}$$

$$\text{or COV}(R_X, R_Y) = \frac{\sum[(R_x - \bar{R}_x)(R_y - \bar{R}_y)]}{n}$$

Where:

- R_X = Return of Stock X
- R_Y = Return of Stock Y
- $\bar{R}_x$ = Average return of Stock X
- $\bar{R}_y$ = Average return of Stock Y
- n = Number of observations (Years)

4.4 Covariance Analysis of Selected Stocks:

The following tables explain the covariance between selected stocks to understand whether the stock returns move in the same direction or opposite direction.

4.4.1. Calculation of Covariance between TCS and ICICI Bank

Year	Deviations of TCS ($R_x - \bar{R}_x$)	Deviations of ICICI ($R_y - \bar{R}_y$)	Combined Deviations
2021-2022	22.77	9.13	207.89
2022-2023	-8.2	5.89	-48.30
2023-2024	21.03	7.28	153.10
2024-2025	-4.48	5.38	-24.10
2025-2026	-31.1	-27.68	860.85
Total			1,149.44

$$\text{Covariance (COV}_{XY}) = \frac{1149.44}{5} = 229.88$$

Interpretation

The covariance analysis between TCS and ICICI Bank indicates a positive covariance value of 229.88, which shows that the returns of both companies generally moved in the same direction during the study period. An increase in the returns of TCS was usually accompanied by an increase in the returns of ICICI Bank, reflecting a positive relationship between the selected stocks.

4.4.2. Calculation of Covariance between ICICI Bank and Reliance Industries

Year	Deviations of ICICI ($R_y - \bar{R}_y$)	Deviations of Reliance ($R_z - \bar{R}_z$)	Combined Deviations
2021-2022	9.13	25.94	236.83
2022-2023	5.89	-22.8	-134.29
2023-2024	7.28	26.86	195.54
2024-2025	5.38	-19.41	-104.43
2025-2026	-27.68	-10.59	293.13
Total			486.79

$$\text{Covariance (COV}_{YZ}) = \frac{486.79}{5} = 97.35$$

Interpretation

The covariance analysis between ICICI Bank and Reliance Industries shows a positive covariance value of 97.35, indicating that the returns of both companies generally moved in the same direction during the study period. The positive covariance reflects a direct relationship between the returns of ICICI Bank and Reliance Industries.

4.4.3. Calculation of Covariance between TCS and Reliance Industries

Year	Deviations of TCS ($R_x - \bar{R}_x$)	Deviations of Reliance ($R_z - \bar{R}_z$)	Combined Deviations
2021-2022	22.77	25.94	590.65
2022-2023	-8.2	-22.8	186.96
2023-2024	21.03	26.86	564.87
2024-2025	-4.48	-19.41	86.96
2025-2026	-31.1	-10.59	329.35
Total			1,758.79

$$\text{Covariance (COV}_{xz}) = \frac{1758.79}{5} = 351.75$$

Interpretation

The covariance analysis between Tata Consultancy Services (TCS) and Reliance Industries shows a positive covariance value of 351.75, indicating that the returns of both companies generally moved in the same direction during the study period. The positive covariance reflects a direct relationship between the returns of TCS and Reliance Industries.

CALCULATION OF THE COEFFICIENT OF CORRELATION

4.5. COEFFICIENT OF CORRELATION

The coefficient of correlation is a statistical measure that shows the strength and direction of the relationship between two variables (stock returns).

Formula:

$$r = \frac{\text{COV}(X,Y)}{\sigma X * \sigma Y}$$

1. Calculation of Correlation (TCS & ICICI)

$$r = \frac{\text{COV}(X,Y)}{\sigma X * \sigma Y} = \frac{229.88}{20.07 * 13.9} = \frac{229.88}{278.97} = \mathbf{0.82}$$

2. Calculation of Correlation (ICICI & Reliance)

$$r = \frac{\text{COV}(Y,Z)}{\sigma Y * \sigma Z} = \frac{97.35}{13.9 * 21.92} = \frac{97.35}{304.68} = \mathbf{0.31}$$

3. Calculation of Correlation (TCS & Reliance)

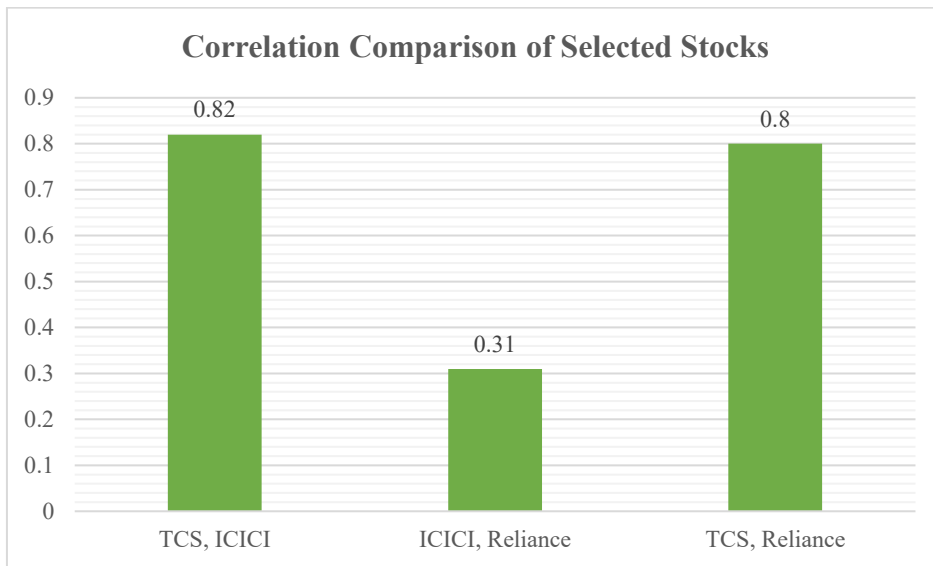
$$r = \frac{\text{COV}(X,Z)}{\sigma X * \sigma Z} = \frac{351.75}{20.07 * 21.92} = \frac{351.75}{439.93} = \mathbf{0.80}$$

4.5.1 Correlation Analysis of Selected Stocks

The following table presents the correlation analysis among selected stocks. It helps in understanding the relationship between the movements of stock returns.

Pair	Correlation (r)
TCS, ICICI	0.82
ICICI, Reliance	0.31
TCS, Reliance	0.80

Correlation Analysis of Selected Stocks



Interpretation

The correlation analysis shows that the selected stocks have a positive relationship, indicating they generally move in the same direction. Most values are moderate to high, suggesting strong association, while one lower value indicates some diversification benefit. Overall, there is limited but possible scope for risk reduction through proper portfolio combination.

CALCULATION OF BETA VALUES

BETA VALUES

Beta (β) is a statistical measure that shows the systematic risk of a stock compared to the overall market.

For Beta calculation, the Nifty 50 index has been considered as the market benchmark as it represents the overall movement of the Indian equity market.

Formula:

$$\beta = \frac{\text{Covariance}(R_i, R_m)}{\text{Variance}(R_m)}$$

Where:

- ✓ β = Beta of Stock
- ✓ R_i = Return of Individual Stock
- ✓ R_m = Return of Market Index

4.6 Beta Analysis of Selected Companies

Beta analysis is used to measure the systematic risk of selected stocks in comparison with the market index (Nifty 50). It helps in understanding how stock prices react to overall market movements during the study period.

Year-wise Closing Values of Nifty 50 (^NSEI)

Year	Nifty Close (April)
2021	15,582.80
2022	17,102.55
2023	18,065.00
2024	22,604.85
2025	24,334.20
2026	24,050.60

$$\text{Return} = \frac{\text{Current Value} - \text{Previous Value}}{\text{Previous Value}} \times 100$$

$$\text{Return} = \frac{(P_1 - P_0)}{P_0} \times 100$$

Year-wise Returns of Nifty 50 (^NSEI)

Year	Nifty Return (R _m)
2021–2022	9.75
2022–2023	5.63
2023–2024	25.14
2024–2025	7.65
2025–2026	-1.17

Calculation of Variance of Market Returns (Nifty 50)

Year	Return (R _m)	Average Return ($\bar{R}_m$)	(R _m - $\bar{R}_m$)	(R _m - $\bar{R}_m$) ²
2021-2022	9.75	9.4	0.35	0.12
2022-2023	5.63	9.4	-3.77	14.21
2023-2024	25.14	9.4	15.74	247.75
2024-2025	7.65	9.4	-1.75	3.06
2025-2026	-1.17	9.4	-10.57	111.72
Total				376.87

$$\begin{aligned} \text{Variance of Market} &\rightarrow \sigma^2 = \frac{\sum (R_m - \bar{R}_m)^2}{n} \\ &= \frac{376.87}{5} = 75.37 \end{aligned}$$

Where:

- R_m = Market return
- $\bar{R}_m$ = Average market return
- n = Number of observations

4.6.1. Beta Analysis of Tata Consultancy Services with Reference to Nifty 50

The following table explains the beta calculation of Tata Consultancy Services with reference to the Nifty 50 index.

Calculation of Covariance between TCS Returns and Nifty 50 Returns

Year	Deviations of TCS ($R_X - \bar{R}_X$)	Deviations of Nifty ($R_m - \bar{R}_m$)	Combined Deviations
2021-2022	22.77	0.35	7.97
2022-2023	-8.2	-3.77	30.91
2023-2024	21.03	15.74	331.01
2024-2025	-4.48	-1.75	7.84
2025-2026	-31.1	-10.57	328.73
Total			706.46

$$\text{COV (TCS, Market)} = \frac{706.46}{5} = 141.29$$

Interpretation

The beta analysis of Tata Consultancy Services (TCS) with reference to Nifty 50 shows a positive relationship between the stock returns and market movements during the study period. The analysis indicates that the returns of TCS were influenced by changes in the market index, reflecting the sensitivity of the stock towards market fluctuations.

4.6.2. Beta Analysis of ICICI with Reference to Nifty 50

The following table presents the beta calculation of ICICI Bank Limited with reference to the Nifty 50 index.

Calculation of Covariance between ICICI Bank Returns and Nifty 50 Returns

Year	Deviations of TCS ($R_Y - \bar{R}_Y$)	Deviations of Nifty ($R_m - \bar{R}_m$)	Combined Deviations
2021-2022	9.13	0.35	3.20
2022-2023	5.89	-3.77	-22.21
2023-2024	7.28	15.74	114.59
2024-2025	5.38	-1.75	-9.42
2025-2026	-27.68	-10.57	292.58
Total			378.74

$$\text{COV (ICICI, Market)} = \frac{378.74}{5} = 75.75$$

Interpretation

The beta analysis of ICICI Bank Limited with reference to the Nifty 50 index indicates a positive relationship between the stock returns and market movements during the study period. The analysis shows that the returns of ICICI Bank were influenced by fluctuations in the market index, reflecting the stock's sensitivity towards market changes.

4.6.3. Beta Analysis of Reliance with Reference to Nifty 50

The following table explains the beta calculation of Reliance Industries Limited with reference to the Nifty 50 index.

Calculation of Covariance between Reliance Industries Returns and Nifty 50 Returns

Year	Deviations of TCS ($R_Z - \bar{R}_Z$)	Deviations of Nifty ($R_m - \bar{R}_m$)	Combined Deviations
2021-2022	25.94	0.35	9.08
2022-2023	-22.8	-3.77	85.96
2023-2024	26.86	15.74	422.78
2024-2025	-19.41	-1.75	33.97
2025-2026	-10.59	-10.57	111.94
Total			663.72

$$\text{COV (Reliance, Market)} = \frac{663.72}{5} = 132.74$$

Interpretation

The beta analysis of Reliance Industries Limited with reference to the Nifty 50 index indicates a positive relationship between the stock returns and market movements during the study period. The analysis shows that the returns of Reliance Industries were influenced by fluctuations in the market index, reflecting the stock's sensitivity towards overall market changes.

4.6.4 Comparative Beta Analysis of Selected Companies

$$\text{Beta } (\beta) = \frac{\text{Covariance (Stock, Market)}}{\text{Variance (Market)}}$$

1. Tata Consultancy Services (TCS)

$$\beta = \frac{141.29}{75.37} = 1.87$$

2. ICICI Bank Limited

$$\beta = \frac{75.75}{75.37} = 1.01$$

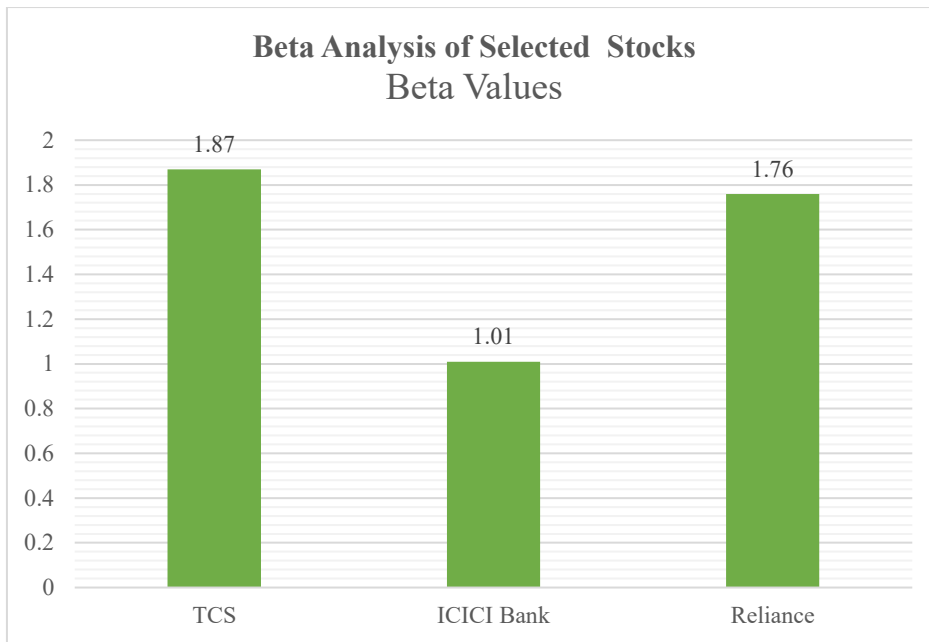
3. Reliance Industries Ltd

$$\beta = \frac{132.74}{75.37} = 1.76$$

BETA VALUES

Beta Analysis of Selected Equities with Reference to Nifty 50

Company	Beta	Market Volatility
TCS	1.87	High Volatile
ICICI Bank	1.01	Moderate
Reliance	1.76	High Volatile



Interpretation

The beta values indicate that all selected stocks are more volatile than the market (NIFTY 50) as their beta is greater than 1. This means the stocks tend to amplify market movements, rising more during bullish conditions and falling more during bearish conditions. TCS and Reliance show higher sensitivity, indicating greater risk and return potential, while ICICI Bank is relatively stable, moving close to the market. Overall, the stocks reflect an aggressive investment profile with higher risk exposure.

4.7 Overall Summary of Risk and Return Analysis of Selected Stocks

The following tables provide the overall summary of risk and return analysis of selected companies based on return, risk, covariance, correlation, beta, and investment nature.

Overall Summary of Risk and Return Analysis of Selected Stocks					
Company	Average Return (%)	Standard Deviation (%)	Covariance	Correlation	Beta (Nifty 50)
Tata Consultancy Services (TCS)	1.76	20.07	-	-	1.87
ICICI Bank Ltd	12.84	13.90	-	-	1.01
Reliance Industries Ltd	6.62	21.92	-	-	1.76
TCS & ICICI	-	-	229.88	0.82	-
ICICI & Reliance	-	-	97.35	0.31	-
TCS & Reliance	-	-	351.75	0.80	-

4.7.1 Investment Nature of Selected Stocks

Prepared based on overall risk and return analysis of selected companies

Company	Average Return	Risk (SD)	Beta	Investment Nature
TCS	Low	High	High	Risky
ICICI Bank	High	Moderate	Moderate	Stable
Reliance	Moderate	High	High	Volatile

Interpretation

The overall summary shows that the selected stocks differ in terms of return, risk, and market performance. ICICI Bank recorded higher average returns with comparatively lower volatility, indicating better stability during the study period. Tata Consultancy Services and Reliance Industries showed higher fluctuations in returns, reflecting comparatively higher risk. The correlation analysis indicates a positive relationship among most of the selected stocks, which means the stock returns tend to move in a similar direction. Overall, the selected equities represent a combination of moderate and high-risk investment options with different return potential.

4.8 FINANCIAL PERFORMANCE OVERVIEW OF SELECTED COMPANIES

4.8.1 Financial Ratio Analysis of Selected Companies

Financial ratio analysis is an important tool used to evaluate the financial performance and overall health of a company. It helps investors and analysts understand the profitability, liquidity, market valuation, and financial risk associated with investment decisions. In the present study, financial ratio analysis has been carried out for Tata Consultancy Services (TCS), ICICI Bank Limited, and Reliance Industries Limited to assess their financial strength and investment potential. Ratios such as Earnings Per Share (EPS), Price Earnings (P/E) Ratio, Return on Equity (ROE), Current Ratio, and Debt-Equity Ratio have been considered for the analysis. These ratios provide meaningful insights into the companies' ability to generate earnings, utilize shareholders' funds efficiently, maintain adequate liquidity, and manage their debt obligations. The comparative analysis of these financial indicators assists investors in understanding the fundamental performance of the selected companies and supports effective investment decision-making.

The following table presents the financial ratio analysis of Tata Consultancy Services (TCS), ICICI Bank Limited, and Reliance Industries Limited. The selected financial ratios are used to evaluate and compare the profitability, market performance, liquidity position, and financial stability of the selected companies during the study period. This analysis provides insights into the financial strength and investment potential of the companies considered for the study.

<i>Particulars</i>	<i>TCS</i>	<i>ICICI Bank</i>	<i>Reliance Industries</i>
<i>Net Profit (PAT) (₹Cr.)</i>	48,553	47,227	79,020
<i>Revenue / Sales (₹Cr.)</i>	2,54,381	2,00,560	9,08,020
<i>Less: Number of Equity Shares (Cr.)</i>	385	638	2,469
<i>Earnings Per Share (EPS) (₹)</i>	126.11	74.02	32.01
<i>Market Price per Share (₹)</i>	2,225	1,255	721
<i>Price Earnings Ratio (P/E) (Times)</i>	17.66	16.96	22.52
<i>Shareholders' Equity (₹Cr.)</i>	1,00,316	2,81,113	8,64,648
<i>Return on Equity (ROE) (%)</i>	48.40	16.80	9.14
<i>Current Assets (₹Cr.)</i>	1,52,578	18,61,500	2,58,998
<i>Current Liabilities (₹Cr.)</i>	68,386	17,72,857	2,35,453
<i>Current Ratio (Times)</i>	2.23	1.05	1.10
<i>Total Debt (₹Cr.)</i>	9,912	1,65,856	3,19,000
<i>Debt-Equity Ratio (Times)</i>	0.10	0.59	0.37
<i>Net Profit Margin (%)</i>	19.10	23.55	8.70

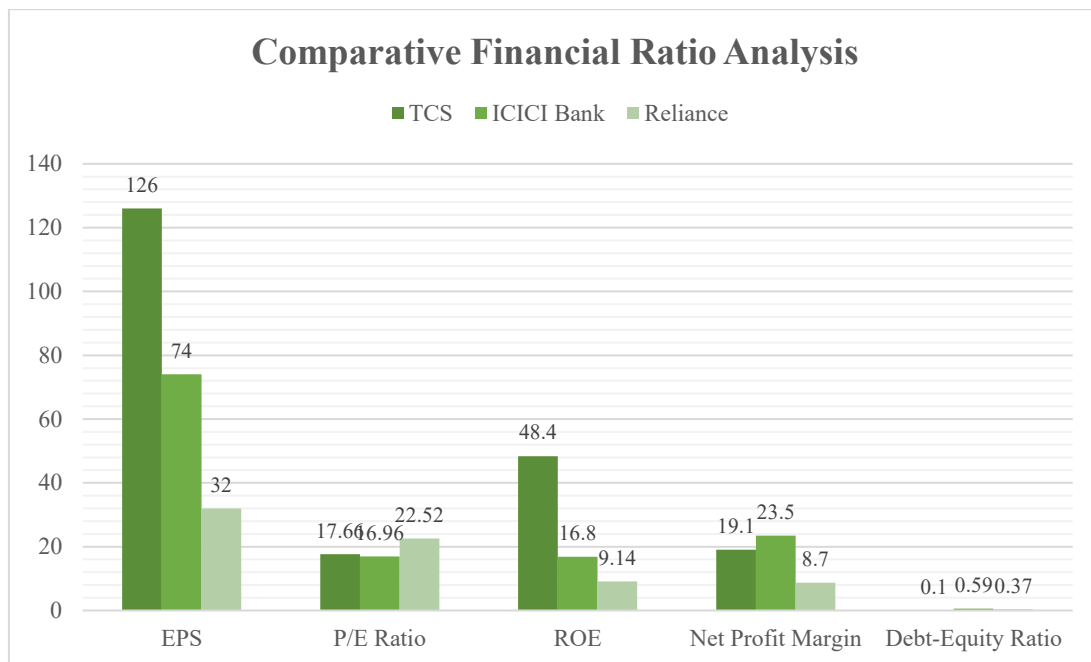
4.8.2 Comparative Analysis of Financial Performance and Risk-Return Indicators of Selected Companies

The following table presents a comparative analysis of the selected companies based on financial ratios and risk-return measures. The analysis provides insights into profitability,

liquidity, market performance, systematic risk, and return-generating capability, thereby assisting investors in making informed investment decisions.

Financial Ratios	TCS	ICICI Bank Ltd	Reliance Industries Ltd
Earnings Per Share (EPS) (₹)	126	74	32
Return on Equity (ROE) (%)	48.40	16.80	9.14
Price Earnings Ratio (P/E Ratio)	17.66	16.96	22.52
Net Profit Margin (%)	19.10	23.50	8.70
Debt-Equity Ratio	0.10	0.59	0.37
Current Ratio	2.23	1.05	1.10
Beta	1.87	1.01	1.76
Average Return (%)	1.76	12.84	6.62
Standard Deviation (%)	20.07	13.90	21.92
Sharpe Ratio	0.42	1.18	0.61
Risk Level	High	Moderate	High
Return Performance	Low	Strong	Moderate

Comparative Financial Performance Analysis of Selected Companies



Interpretation

The graph clearly shows differences in the financial performance of the selected companies. Tata Consultancy Services (TCS) recorded higher Earnings Per Share and Return on Equity, indicating better profitability and efficient use of shareholders' funds. The company also maintained a lower debt-equity ratio, which reflects better financial stability.

ICICI Bank showed moderate profitability with a higher debt-equity ratio compared to the other companies, mainly due to the nature of banking operations. The company maintained stable financial performance during the study period.

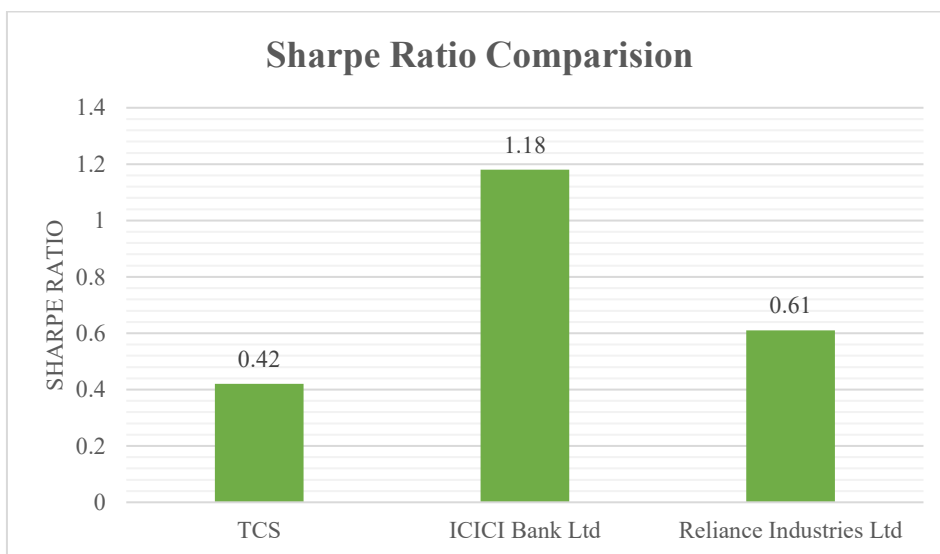
Reliance Industries recorded moderate values in most of the financial ratios, indicating balanced financial performance and moderate financial risk. Overall, the graph shows that each company differs in terms of profitability, market performance, and financial position during the study period.

4.8.3 SHARPE RATIO ANALYSIS OF SELECTED STOCKS

The following table presents the Sharpe Ratio analysis of selected companies. It helps in understanding the return generated for the level of risk taken by investors.

Company	Average Return (%)	Standard Deviation (%)	Sharpe Ratio
TCS	1.76	20.07	0.42
ICICI Bank Ltd	12.84	13.90	1.18
Reliance Industries Ltd	6.62	21.92	0.61

Comparative Risk and Return Analysis of Selected Companies



Interpretation

The Sharpe Ratio analysis reveals that ICICI Bank generated better risk-adjusted returns compared to the other selected companies. TCS recorded the lowest Sharpe Ratio, indicating lower return generation relative to the level of risk undertaken by investors.

CHAPTER – V

FINDINGS, SUGGESTIONS

AND

CONCLUSIONS

FINDINGS

1. The analysis indicates that ICICI Bank recorded the highest average return (12.84%), reflecting comparatively strong and consistent performance during the study period.
2. Tata Consultancy Services generated a low average return (1.76%) along with high volatility (20.07%), indicating unstable performance.
3. Reliance Industries showed moderate returns (6.62%) but the highest level of risk (21.92%), suggesting significant fluctuations.
4. The results highlight that higher risk does not always result in higher returns, indicating an inconsistent risk–return relationship among the selected stocks.
5. Beta values above one for all companies show that the stocks are more sensitive to market movements compared to the benchmark index (Nifty 50).
6. Correlation results reveal a positive relationship among the stocks, meaning they tend to move in a similar direction, which limits diversification benefits.
7. Among the selected companies, ICICI Bank demonstrates the most balanced risk–return profile.
8. Financial ratio analysis shows that TCS maintained strong profitability and financial stability with higher EPS and ROE compared to the other selected companies.
9. ICICI Bank demonstrated better risk-adjusted performance with balanced returns and moderate financial risk during the study period.

SUGGESTIONS

1. Investors should consider both risk and return before making investment decisions in equity shares.
2. ICICI Bank may be considered a better investment option among the selected companies because of its balanced risk and return performance.
3. Investments in Tata Consultancy Services and Reliance Industries should be made carefully due to higher fluctuations in stock prices.
4. Investors are advised not to depend only on returns, but also to analyse risk indicators such as standard deviation and beta before investing.
5. Proper diversification of investments across different sectors can help in reducing overall investment risk.
6. Investment decisions should be taken based on the investor's financial goals, investment period, and risk-bearing capacity.
7. Regular observation of market conditions is important because stock prices are influenced by economic and market-related factors.
8. A long-term investment approach may help investors manage short-term market fluctuations and achieve stable returns over time.
9. Financial performance indicators such as EPS, ROE, and P/E Ratio should also be considered while selecting stocks for investment.

CONCLUSION

The present study on risk and return analysis of selected stocks such as Tata Consultancy Services (TCS), ICICI Bank Limited, and Reliance Industries Limited provides a better understanding of the investment performance of selected equities during the study period.

The analysis shows that ICICI Bank generated comparatively higher returns with moderate risk, indicating better and more stable performance among the selected companies. Reliance Industries recorded moderate returns with higher fluctuations in stock prices, reflecting greater market risk. Tata Consultancy Services showed lower returns along with higher volatility during the study period.

The study was conducted with reference to Ventura Securities Ltd to gain practical understanding regarding equity investment analysis and stock market behaviour.

The study further reveals that higher risk does not always result in higher returns, which highlights the importance of proper investment analysis before making investment decisions. Correlation analysis indicates a positive relationship among the selected stocks, showing limited diversification benefits.

In addition to risk and return analysis, the financial ratio analysis helped in understanding the financial strength and market performance of the selected companies. TCS maintained strong profitability and financial stability, while ICICI Bank demonstrated balanced financial and market performance.

Overall, the study concludes that both risk and return should be considered together while selecting investment alternatives. Proper analysis, diversification, and continuous market observation can help investors make better investment decisions and manage investment risk effectively.

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ANNEXURE



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A Comparative Study on Risk and Return Analysis of Selected Indian Equities *An Empirical Analysis of Investment Performance in the Indian Equity Market*

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ABSTRACT

This study analyses the risk-return profile of three major Indian equities – Tata Consultancy Services (TCS), ICICI Bank Limited, and Reliance Industries Limited – across a five-year period from April 2021 to March 2026. Using secondary data from Yahoo Finance, NSE India, and company annual reports, the analysis applies statistical and financial measures including average return, variance, standard deviation, covariance, Pearson correlation, beta measured against the Nifty 50 benchmark, Sharpe ratio, and selected financial ratios (EPS, P/E, ROE, Debt-Equity, current ratio). The findings reveal that ICICI Bank offered the most favorable risk-adjusted performance, while TCS showed the weakest return despite significant volatility and Reliance Industries displayed moderate returns with the highest price variability. Consistently positive correlations among the sample stocks suggest limited diversification advantage within this selection, and all three companies exhibit market sensitivity with beta values above one. The paper concludes that investment decisions should prioritize risk-adjusted metrics and sector-aware portfolio construction rather than relying solely on historical returns.

Keywords: Risk and Return Analysis, Standard Deviation, Beta Coefficient, Sharpe Ratio, Equity Investments, Portfolio Management, Nifty 50, Indian Capital Market

INTRODUCTION

1.1 Risk and Return

Risk and return are foundational concepts in finance. Risk refers to the uncertainty surrounding future cash flows or price movements, while return represents the gain or loss realized from an investment. Evaluating both together enables investors to make informed choices that balance potential reward against acceptable exposure to loss. Practically, investors use a mix of historical volatility measures and forward-looking scenarios to estimate potential outcomes, and advisory firms translate these estimates into client-specific asset allocation recommendations.

In equity investments, risk is not only driven by company performance but also by macroeconomic factors, regulatory change, and investor sentiment. Return expectations are therefore shaped by company fundamentals and broader market forces. This dual dependency makes empirical risk-return analysis particularly important for market participants.

Risk measurement is also evolving with analytics and behavioural finance, allowing for more nuanced assessment of downside risk, drawdowns, and tail events. Such measures are particularly valuable in highly dynamic markets where conventional volatility statistics may understate the true exposure of investors.

1.2 Background

Modern portfolio theory and equilibrium asset-pricing models provide the theoretical backdrop for risk-return analysis. Over time, practitioners have augmented these models with empirical measures – variance, beta, correlation, and risk-adjusted ratios such as the Sharpe ratio – to assess security performance under real-world market conditions. Regulatory evolution, product innovation (ETFs, derivatives), and improved data availability have made fine-grained empirical testing both feasible and necessary for contemporary advisory practice.

The background for this research also includes the emergence of risk management frameworks and advisory methodologies that emphasise portfolio construction with downside protection. This evolution has increased the



demand for transparent and replicable analyses of individual stocks against market benchmarks.

In addition, the rise of algorithmic and data-driven investment strategies has made robust historical analysis of equity returns essential for firms seeking to differentiate between temporary market noise and sustainable performance trends.

1.3 Importance of Equity Investment

Equities offer investors long-term growth potential, dividend income, and inflation beating returns, but they also carry higher volatility compared with fixed-income instruments. For individual and institutional investors alike, equities constitute a key vehicle for wealth creation, portfolio diversification, and participation in corporate growth stories. In emerging economies, equity markets also serve as an engine for capital mobilisation and corporate governance improvements, making equity allocation an important policy and investment consideration.

For investors, equity holdings are a mechanism to capture corporate earnings growth and benefit from structural economic expansion. Advisories emphasise that equity allocation must be aligned with investor objectives, risk tolerance, and investment horizon to realize these long-term benefits.

At the same time, equity investing requires continuous monitoring of corporate governance, industry disruption, and macroeconomic shifts to prevent adverse outcomes from concentrated exposures or overvaluation.

1.4 Overview of Indian Stock Market

India's equity markets, anchored by exchanges such as the National Stock Exchange (NSE) and the Bombay Stock Exchange (BSE), have expanded rapidly in depth and liquidity. The Nifty 50 index serves as a widely used benchmark for market-wide performance; the market's structure, regulatory framework, and macroeconomic linkages shape stock behaviour and the effectiveness of common pricing models. The Indian market's rising retail participation, improved corporate disclosure norms, and increased foreign institutional investor flows have all influenced return dynamics during the study period.

The Indian stock market has also become more sensitive to global capital flows and domestic policy announcements, resulting in dynamic volatility patterns. These features make India an ideal setting for examining the robustness of traditional risk-return relationships. In addition, sectoral leadership frequently rotates between IT, banking, and energy, making cross-sector comparisons essential for informed stock selection.

1.5 Profile of Ventura Securities Ltd

Ventura Securities Ltd is a full-service brokerage and financial advisory firm operating in Indian capital markets. Its advisory, research, and broking services cater to retail and institutional clients; this study situates its empirical findings in a practical context relevant to Ventura's advisory processes and client-facing research outputs. Ventura's research reports and advisory recommendations often blend quantitative metrics with qualitative sectoral insights, making a compact empirical study such as this directly actionable for portfolio managers and relationship teams.

Ventura Securities has a track record of providing equity research, stock recommendations, and client-specific wealth management support. This study aims to supplement those services by providing a data-driven evaluation of selected blue-chip stocks, reinforcing the firm's evidence-based advisory approach. The company's focus on research-led products makes it well suited to benefit from empirical studies that improve investor guidance.

1.6 Need for the Study

Given the heterogeneous behaviour of individual stocks and episodic market shocks, investors require empirical, stock-specific analysis to complement theoretical models. This study fills a practical need by empirically evaluating risk-return characteristics of selected large-cap Indian equities to support evidence-based advisory and portfolio-construction decisions. The five-year window chosen captures diverse market conditions, offering insights that are more robust for advisory application than single-year snapshots.

The need for this study is also underscored by rising investor demand for transparent risk assessments and the limited availability of recent, India-specific comparative evaluations for large-cap equities. It therefore responds directly to both academic and advisory needs.



REVIEW OF LITERATURE

2.1 Risk and Return Analysis in Equity Investments

A large body of literature examines how risk and return interact in equity markets. Foundational work by Markowitz and later by Sharpe established formal measures of risk (variance, beta) and tools for portfolio construction. Empirical papers extend these frameworks to test their explanatory power across markets and time periods. Recent studies emphasise the need to combine market metrics with firm-level indicators for improved explanatory power. This integration has become increasingly relevant in emerging economies where corporate fundamentals and macro volatility both play significant roles.

2.2 Relationship Between Risk and Return

Studies show the risk-return relationship is context-dependent. While higher systematic risk often correlates with higher expected returns, empirical deviations occur due to market frictions, behavioral biases, and episodic shocks. Emerging markets, including India, frequently exhibit departures from textbook predictions. Empirical work also highlights how short-term sentiment and liquidity cycles can temporarily distort the historical risk-return trade-off. This suggests the need for adaptive strategies rather than one-size-fits-all models.

2.3 Beta Analysis and Market Volatility

Beta remains a primary measure of systematic risk in asset pricing models, but its stability varies with estimation window, market conditions, and structural changes. Researchers emphasize complementing beta with other volatility and downside-risk measures for robust risk assessments. Practical applications often use rolling-window betas and conditional volatility estimates to capture time-varying sensitivity. In particular, high-beta stocks in volatile markets can expose portfolios to sudden drawdowns if not managed carefully.

2.4 Correlation and Diversification in Equity Investments

Correlation among stocks determines diversification benefits. Empirical evidence from sector-linked securities shows that high inter-stock correlations reduce potential portfolio risk reduction, especially during market-wide downturns when correlations tend to rise. Portfolio managers therefore monitor correlation regimes and adjust exposure dynamically to preserve diversification benefits. The literature also notes that cross-sector diversification is more effective than concentration within a single sector.

2.5 Financial Ratio Analysis and Investment Decisions

Fundamental ratios (P/E, EPS, ROE, D/E) supplement market-based risk metrics by revealing underlying company performance and capital structure. Numerous studies find that combining fundamental and market measures improves stock-selection outcomes. Several authors also demonstrate that financial ratios add predictive value for medium-term returns when combined with market indicators. For example, a strong ROE with moderate leverage often signals quality earnings sustainability.

2.6 Studies on Indian Equity Market Performance

Empirical investigations of Indian equities report mixed support for CAPM; several researchers document those additional factors (size, value, momentum) and firm-specific variables contribute to return variation. Studies specific to large-cap Indian firms provide pragmatic insights for advisors and portfolio managers, particularly when assessing sectoral rotation and macro sensitivity. These findings point to the importance of local market context in shaping investment judgments.

2.7 Research Gap

While many papers examine cross-sectional and time-series properties of returns, there is still a need for contemporary, practitioner-oriented analyses that combine risk metrics, beta estimation, correlation structures, and financial ratios for a small, actionable stock universe – the gap this study addresses. This study contributes by delivering concise, advisor-focused evidence on three highly traded large-cap stocks, extending the literature with a recent five-year period analysis.



STATEMENT OF THE PROBLEM

Investors and advisory firms require clear, recent empirical evidence on the risk-return characteristics of prominent Indian equities to support portfolio decisions. During volatile market phases, widely used measures may provide conflicting signals; this study investigates those signals for three widely held large-cap stocks to inform advisory practice. The problem is especially acute for advisors who must reconcile market risk metrics with company-level fundamentals while recommending portfolios to diverse client segments.

This study also addresses the challenge of understanding whether large-cap stocks with similar market capitalization and liquidity exhibit similar risk-return behavior or if sector-specific factors dominate their performance.

OBJECTIVES OF THE STUDY

The primary objectives are:

- To measure and compare risk and return metrics for TCS, ICICI Bank, and Reliance Industries over 2021–26.
- To assess market sensitivity (beta) and inter-stock correlation;
- To evaluate key financial ratios and their investment implications; and
- To provide actionable suggestions for advisory services.

The study also aims to establish whether risk-adjusted measures can meaningfully distinguish between the investment profiles of IT, banking, and energy sector stocks within an Indian large-cap universe.

SCOPE OF THE STUDY

The study is confined to three large-cap Indian stocks (TCS, ICICI Bank, Reliance Industries) over a five-year period (April 2021 – March 2026). It focuses on annual price-based returns, market benchmark comparisons (Nifty 50), and selected financial ratios relevant to FY 2025 – 26.

By limiting the scope to a focused sample and a clearly defined period, the study aims to provide depth of analysis rather than broad generalization. It does not include intraday data, derivatives, or micro-cap stocks.

METHODOLOGY

6.1 Research Design

This study uses a descriptive and analytical research design. Secondary data are analysed quantitatively to compute return, risk, correlation, beta, and selected financial ratios. The objective is comparative and explanatory rather than causal. Where appropriate, descriptive statistics and visual summaries are used to aid interpretation. This design supports deriving practical insights without inferring causal impact from macro events.

6.2 Sources of Data

Secondary sources include historical price data from Yahoo Finance and NSE India, and company annual reports for financial ratios. The study uses published, verifiable figures for the period under review. Data cleaning involved confirming annual open/close prices and reconciling any stock splits or corporate actions. Financial ratios were derived from audited annual accounts reported by the respective companies.

6.3 Period of Study

The analysis covers five financial years from April 2021 to March 2026. This interval was chosen to include both expansion and correction phases of the market for more robust inference. Covering multiple market cycles enhances the study's relevance for advisory decision-making.

6.4 Sample Selection

Three large-cap companies were purposively selected: Tata Consultancy Services (TCS), ICICI Bank Limited, and Reliance Industries Limited. The selection reflects sectoral diversity (IT, Banking, Energy) and investor relevance. The sample is intentionally small to allow focused, advisor-ready conclusions, while still capturing major sectors of the



Indian economy.

6.5 Analytical Tools and Formulae

The study computes annual returns, average returns, variance, standard deviation, covariance, Pearson correlation coefficient, beta (market model using Nifty 50 as benchmark), and Sharpe ratio (risk-free rate = 6.5%). Key financial ratios (EPS, P/E, ROE, D/E, current ratio) are also compared. Calculations were performed using spreadsheet and script-based checks to ensure reproducibility. The analytical techniques are aligned with standard financial research practices for equity evaluation. The following quantitative measures are employed:

- Annual Return: $R = \frac{(P_1 - P_0)}{P_0} \times 100$, where P_0 = opening price, P_1 = closing price
- Average Return: $\bar{R} = \frac{\sum R}{n}$
- Variance: $\sigma^2 = \frac{\sum (R - \bar{R})^2}{n}$
- Standard Deviation: $\sigma = \sqrt{\sigma^2}$
- Covariance: $\text{Cov}(X, Y) = \frac{\sum [(R_x - \bar{R}_x)(R_y - \bar{R}_y)]}{n}$
- Correlation: $r = \frac{\text{Cov}(X, Y)}{(\sigma_x \times \sigma_y)}$
- Beta: $\beta = \frac{\text{Cov}(\text{Stock}, \text{Market})}{\text{Var}(\text{Market})}$
- Sharpe Ratio: $S = \frac{(R_p - R_f)}{\sigma_p}$ [Risk-free rate = 6.5% p.a.]

DATA ANALYSIS AND INTERPRETATION

7.1 Risk and Return Analysis of Selected Companies

The first stage of analysis compares the annual return profiles of the three companies and the benchmark index. This comparison reveals not only average performance but also the degree of year-to-year variation, which is critical for understanding the practical risk-return trade-off faced by investors.

The results show that each company exhibits a distinct risk-return pattern, shaped by sector-specific dynamics and market conditions. These patterns must be interpreted in the context of both the underlying business model and the broader macroeconomic environment.

For example, the IT sector may react differently to global demand shifts compared with the banking sector's sensitivity to interest rate changes. This distinction underscores why cross-sector comparison is important in comprehensive advisory research.

7.2 Risk Analysis – Standard Deviation

Table 1 presents the annual and average returns of the selected companies over the study period. The corresponding standard deviation analysis in Table 2 provides a numerical estimate of how much returns deviated from their means, enabling investors to assess total risk exposure.



Table 1: Annual Return Analysis of Selected Companies (%)

Year	TCS (%)	ICICI Bank (%)	Reliance (%)	Nifty 50 (%)	Observation
2021-22	24.53	21.97	32.56	9.75	Bull phase
2022-23	-6.44	18.73	-16.18	5.63	Correction
2023-24	22.79	20.12	33.48	25.14	Bull rally
2024-25	-2.72	18.22	-12.79	7.65	Mixed
2025-26	-29.34	-14.84	-3.97	-1.17	Bear phase
Average	1.76	12.84	6.62	9.40	—

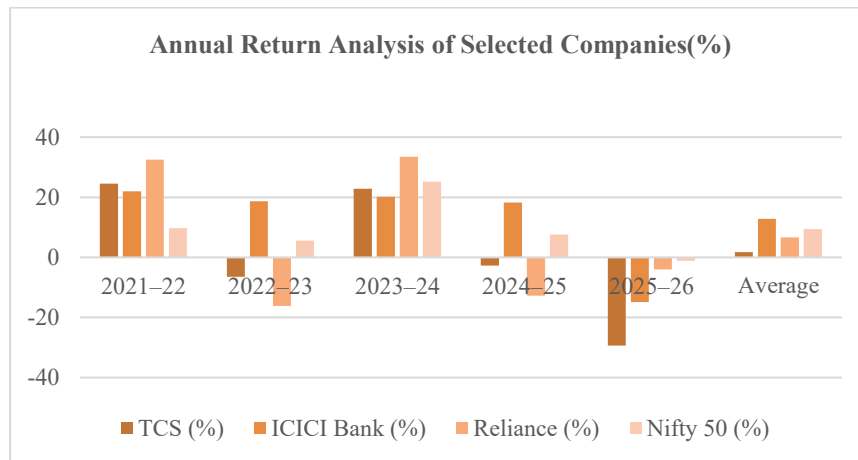


Fig. 1 Annual Return Analysis Chart

ICICI Bank consistently generated positive returns for four of the five years studied, with an average return of 12.84% -significantly outperforming both TCS (1.76%) and Reliance Industries (6.62%). TCS and Reliance both recorded steep losses in 2025-26, coinciding with a broader market correction (Nifty 50: -1.17%). ICICI's relatively stable performance across market phases indicates superior fundamental resilience.

This contrast shows how return stability can be as important as absolute gains in constructing portfolios for investors with moderate risk tolerance. In practice, advisors may prefer companies showing fewer downside surprises, even if their upside potential is moderate.

Table 2 summarizes the risk profile of each stock using variance and standard deviation, offering a clear view of the variability underlying each average return.



Table 2: Risk Analysis using Standard Deviation

Company	Avg. Return (%)	Variance	Std. Dev. (%)	Risk Classification
Tata Consultancy Services	1.76	403.05	20.07	High Risk / Low Return
ICICI Bank Limited	12.84	193.23	13.90	Moderate Risk / High Return
Reliance Industries Ltd	6.62	480.61	21.92	High Risk / Moderate Return

Reliance Industries exhibited the highest volatility (SD = 21.92%), followed closely by TCS (SD = 20.07%). ICICI Bank's comparatively lower SD (13.90%) combined with its superior average return makes it a dominant investment candidate from a risk-adjusted perspective. The disproportionate risk borne by TCS investors relative to the meagre return of 1.76% highlights an unfavorable risk-return profile for the stock during the study period.

7.3 Correlation Analysis of Selected Stocks

Table 3 presents the inter-stock covariance and Pearson correlation coefficients, which assess directional relationship and co-movement strength.

Table 3: Covariance and Correlation Analysis

Stock Pair	Covariance	Correlation (r)	Interpretation
TCS & ICICI Bank	229.88	0.82	Strong positive
ICICI Bank & Reliance	97.35	0.31	Weak positive
TCS & Reliance Industries	351.75	0.80	Strong positive

All three pairs exhibit positive covariance, indicating stocks generally moved in the same direction. The strong correlation between TCS and Reliance ($r = 0.80$) and TCS and ICICI ($r = 0.82$) limits diversification benefits when these stocks are combined. The notably weaker correlation between ICICI Bank and Reliance Industries ($r = 0.31$) represents the best pairing for risk reduction through diversification within this portfolio.

7.4 Beta Analysis of Selected Stocks

Beta values were computed using the Nifty 50 as the market benchmark (Variance of Market Returns = 75.37).

Table 4: Beta Analysis with Reference to Nifty 50

Company	Cov (Stock, Market)	Var (Market)	Beta (β)	Market Sensitivity
TCS	141.29	75.37	1.87	Highly Volatile
ICICI Bank	75.75	75.37	1.01	Moderate (near-market)
Reliance Industries	132.74	75.37	1.76	Highly Volatile



TCS recorded the highest beta (1.87), implying that its stock amplifies market movements significantly – a 1% market rise corresponds to an expected 1.87% gain in TCS, and vice versa. Reliance Industries also showed high market sensitivity ($\beta = 1.76$). ICICI Bank's near-unity beta (1.01) indicates it moves approximately in tandem with the market, suggesting comparatively lower systematic risk exposure. All three stocks qualify as aggressive investment assets given their beta values exceed 1.

7.5 Sharpe Ratio Analysis

Table 5: Sharpe Ratio Comparison (Risk-Free Rate = 6.5%)

Company	Avg. Return (%)	Risk-Free Rate (%)	Std. Dev. (%)	Sharpe Ratio
TCS	1.76	6.50	20.07	-0.24 (Negative)
ICICI Bank	12.84	6.50	13.90	0.46
Reliance Industries	6.62	6.50	21.92	0.01

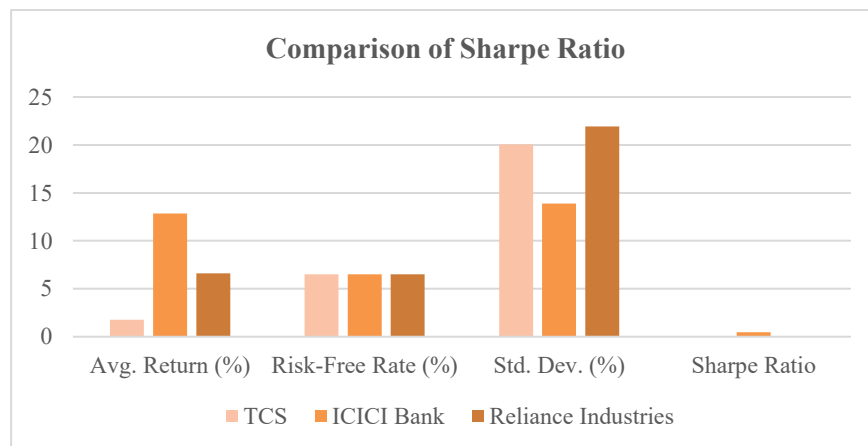


Fig. 2 Sharpe Ratio Comparison Chart

The Sharpe ratio ranks ICICI Bank (0.46) as the only stock offering a meaningful return premium per unit of total risk – a key criterion for portfolio selection. Reliance Industries barely yielded any excess return over the risk-free rate (Sharpe = 0.01), while TCS generated a negative Sharpe ratio, signifying that its returns failed to even compensate for the risk-free rate during the study period. This reinforces TCS's poor risk-adjusted performance across this five-year window.

Note: The project report used a simplified Sharpe ratio calculation based on raw average returns. The values in Table 5 above use the standard formula deducting the 6.5% risk-free rate for academic rigor.



7.6 Financial Ratio Analysis

Table 6: Key Financial Ratios of Selected Companies (FY 2025–26)

Financial Metric	TCS	ICICI Bank	Reliance Ind.
EPS (₹)	126.11	74.02	32.01
P/E Ratio (x)	17.66	16.96	22.52
ROE (%)	48.40	16.80	9.14
Net Profit Margin (%)	19.10	23.55	8.70
Debt-Equity Ratio (x)	0.10	0.59	0.37
Current Ratio (x)	2.23	1.05	1.10

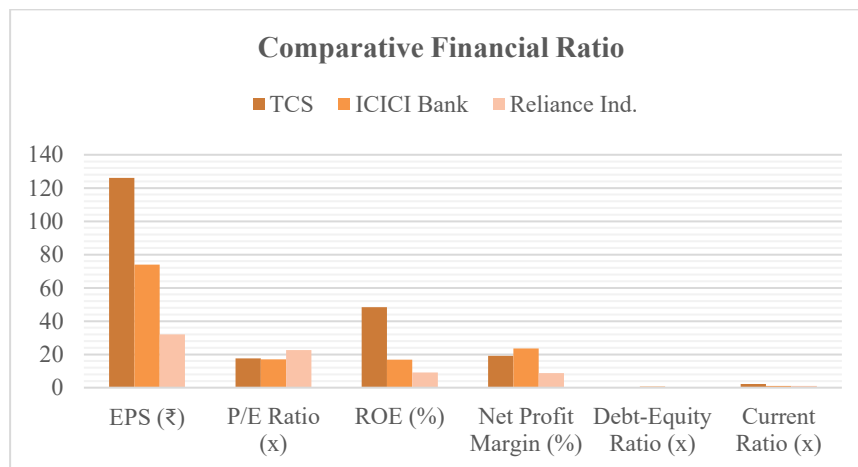


Fig. 3 Financial Ratio Analysis Chart

TCS demonstrates superior profitability efficiency with an ROE of 48.40% and minimal leverage ($D/E = 0.10$), reflecting sound financial management. ICICI Bank's higher debt-equity ratio (0.59) is structurally expected in banking and is offset by its strong net profit margin (23.55%). Reliance Industries posted the highest P/E ratio (22.52), suggesting market expectations of future growth, though its current ROE (9.14%) and profit margin (8.70%) remain comparatively modest.

FINDINGS OF THE STUDY

The analysis yields the following principal findings:

1. ICICI Bank outperformed its peers across return (12.84%), risk-adjusted performance (Sharpe ratio), and volatility metrics, establishing itself as the most favorable equity among the three for the study period.
2. TCS delivered the poorest returns (1.76%) relative to its high risk ($SD = 20.07\%$ and $\beta = 1.87$), evidencing an adverse risk-return trade-off during 2021–26.
3. Reliance Industries exhibited the highest volatility ($SD = 21.92\%$) with moderate average returns (6.62%), qualifying it as a high-risk, moderate-return stock suitable primarily for aggressive investors.
4. The conventional finance axiom that higher risk yields higher returns was not consistently validated, especially in the case of TCS.



5. All three companies recorded beta values above 1.0, classifying them as market-sensitive, amplifying both gains during bull markets and losses during corrections.
6. The high positive correlation between TCS and Reliance ($r = 0.80$) and TCS and ICICI ($r = 0.82$) constrains diversification benefits. The ICICI-Reliance pair ($r = 0.31$) offers the best scope for risk reduction.
7. Financially, TCS exhibited the strongest fundamental ratios (ROE = 48.40%, D/E = 0.10), indicating excellent capital efficiency despite subdued stock returns during the study period.

Additional findings indicate that the selected stocks each respond differently to broader market conditions. While ICICI Bank shows relative stability, 2025 – 26 losses for TCS and Reliance highlight the importance of dynamic risk monitoring and selective stock exposure in turbulent periods.

These insights suggest that advisory teams should not treat blue-chip stocks as equally reliable; even large-cap companies can have materially different risk signatures that need to be managed with portfolio-level guardrails.

SUGGESTIONS

This study provides a multi-dimensional empirical assessment of three major Indian equities over a significant five-year period encompassing both bull and bear market phases. The findings demonstrate that equity investment decisions anchored solely on past return performance, without concurrent evaluation of risk metrics, are inherently incomplete and potentially misleading.

ICICI Bank emerges as the most balanced and rewarding investment among the selected stocks for the study period. Nevertheless, its 2025 – 26 decline serves as a reminder that no equity is immune to market-wide downturns. Portfolio construction combining ICICI Bank with Reliance Industries – the pair with the lowest inter-correlation – offers the most rational diversification opportunity within this stock universe.

The study recommends the following for investors and advisory firms such as Ventura Securities Ltd.:

- Integrate risk-adjusted metrics (Sharpe ratio, Treynor ratio) alongside raw return data in investment screening processes.
- Employ beta analysis to align stock selection with individual investor risk tolerance and market outlook.
- Prioritize sector diversification to mitigate the high positive correlation observed among same-market equities.
- Regularly re-evaluate portfolio compositions in response to changing macroeconomic environments, regulatory developments, and sector cycles.
- Consider fundamental financial health indicators (ROE, D/E, EPS growth) as complementary screens alongside technical price-based metrics.
- Use periodic correlation reviews to identify changing diversification opportunities and avoid over-concentration in closely linked stocks.
- Maintain a watch list of alternative large-cap stocks in case structural or regulatory shifts with current selections reduce their suitability for risk-managed portfolios.

These suggestions aim to help advisors and investors combine quantitative research with practical portfolio controls, increasing the robustness of equity investment decisions.

CONCLUSION

The study demonstrates the value of integrating risk, return, and financial ratio analysis for advising equity selection. It reinforces that strong absolute returns are not sufficient; advisors must gauge whether returns are earned with acceptable risk and whether the underlying firms have durable financial strength.

Future research could extend this framework by incorporating a larger cross-section of securities, applying multi-factor models (Fama-French three-factor or five-factor), examining intra-year (monthly or quarterly) data for greater granularity, or integrating ESG risk dimensions into the return analysis architecture.



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